

MP4201功能简介与应用指南

2025.12

本资料仅供MP4201开源复刻活动使用
请勿外传

MP4201 | 100V、25A、同步双向升降压控制器，带 I2C接口、电池充电和可调节 OCP

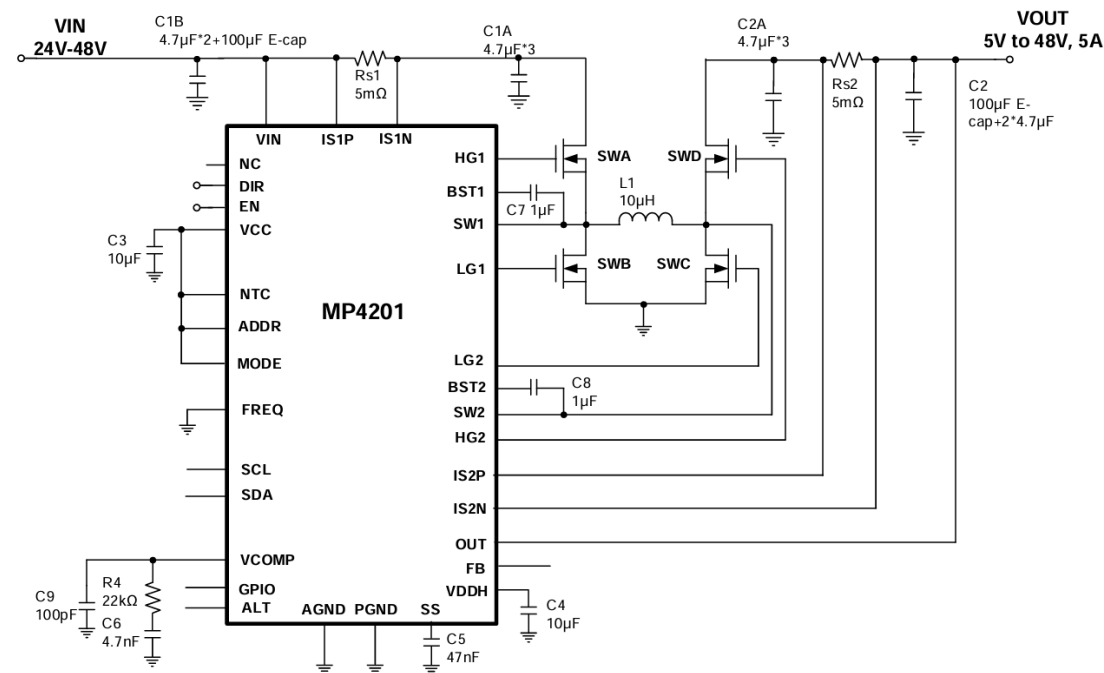
FEATURES

- 4-switch Bi-directional Buck-boost Controller
- 4.75V to 100V Operation Input Voltage Range.
- External feedback can support 100V output; Internal feedback supports 3.2V-80V Output Voltage Range, 20mV/step.
- 1 μ A EN Shutdown IQ
- 100 μ A IQ when I2C operation OFF
- Programmable Input / Output Continuous Current limit (CC), 0.5A-25A with 50mA/step.
- Support 2m Ω /5m Ω Sensing Resistor for 25A Current
- Support USB PD SPR PPS and EPR AVS
- 200kHz/400kHz/600kHz/1MHz Switching Frequency with frequency spread spectrum and frequency SYNC input
- PFM and FCCM mode selection
- Support 5V GATE Driver Compliant with GaN (10V Gate Driver Option)
- 1.75A Peak Source Current and 4.5A Peak Sink Current Capability for Driver
- Programmable VIN UVLO Voltage via OTP
- Programmable VOUT OVP
- Programmable Input Regulation Threshold for MPPT

System

- Battery Charge Cycle Control with Status Indication
- I2C Interface with interrupt function
- OCP, OVP, PG, NTC and thermal shutdown Interrupt
- 10-bit ADC Readout Function for VIN, IIN, VOUT, IOUT and Temperature

SCHEMATIC



APPLICATIONS

- USB EPR Power Delivery
- Solar Power (Battery charger, Power Optimizer)
- Power Tool, Medical

MP4201简介

DESCRIPTION

The MP4201 integrates a synchronous, four-switch N-MOSFET driver supporting GaN as well, Bi-directional buck-boost controller capable of regulating different output voltages with a wide input voltage range and high efficiency.

The MP4201 provides an I²C interface with one-time programmable memory (OTP), which supports V_{OUT} voltage programmability, V_{OUT} slew-rate control, input and output constant current limit programmability, making the MP4201 suitable for USB Standard (SPR), Extended Power Range PD (EPR) design and high power systems.

The MP4201 provides forced continuous conduction mode (FCCM), and Auto PFM/PWM selection mode (PFM), battery charging control and programmable switching frequency with frequency spread spectrum function (FSS), which supports flexible designs for different applications.

The MP4201 integrates Battery charge cycle control with accurate battery voltage regulation, integrate pre-charge, CC/CV, termination charge detection and status indication.

The MP4201 also features programmable input and output over-current protection (OCP) hiccup mode or continuous CC mode without hiccup, short circuits protection, programmable input OVP/UVLO thresholds, output OVP and thermal shut down function (TSD). The NTC pin can be used to sense the PCB board or MOSFET temperature to do over temperature protection.

The MP4201 is available in a QFN-32 (5mmx6mm) package

FEATURES

- 4-switch Bi-directional Buck-boost Controller
- 4.75V⁽¹⁾ to 100V Operation Input Voltage Range.

- External Feedback Can Support 100V Output; Internal Feedback Supports 3.2V-81.92V Output Voltage Range, 20mV/step.
- 1μA EN Shutdown IQ
- 300μA IQ when I2C Operation OFF
- Programmable Input / Output Continuous Current Limit (CC), 0.5A-25A with 50mA/Step.
- Support 2mΩ/5mΩ Programmable Current Sensing Resistor
- Support USB PD SPR PPS and EPR AVS
- 200kHz/400kHz/600kHz/1MHz Switching Frequency with Frequency Spread Spectrum and Frequency SYNC Input
- PFM and FCCM Mode Selection
- Support 5V 1.75A Source/ 4.5A Sink Ability GATE Driver Compliant with GaN
- Programmable VIN UVLO Voltage via OTP
- Programmable VOUT OVP
- Programmable Input Regulation Threshold for MPPT

System

- Battery Charge Cycle Control with Status Indication
- I2C Interface with Interrupt Function.
- OCP, OVP, PG, NTC and Thermal Shutdown Interrupt
- 10-bit ADC Readout Function for VIN, IIN, VOUT, IOUT and Temperature.
- QFN-32 (5mmx6mm) Package

APPLICATIONS

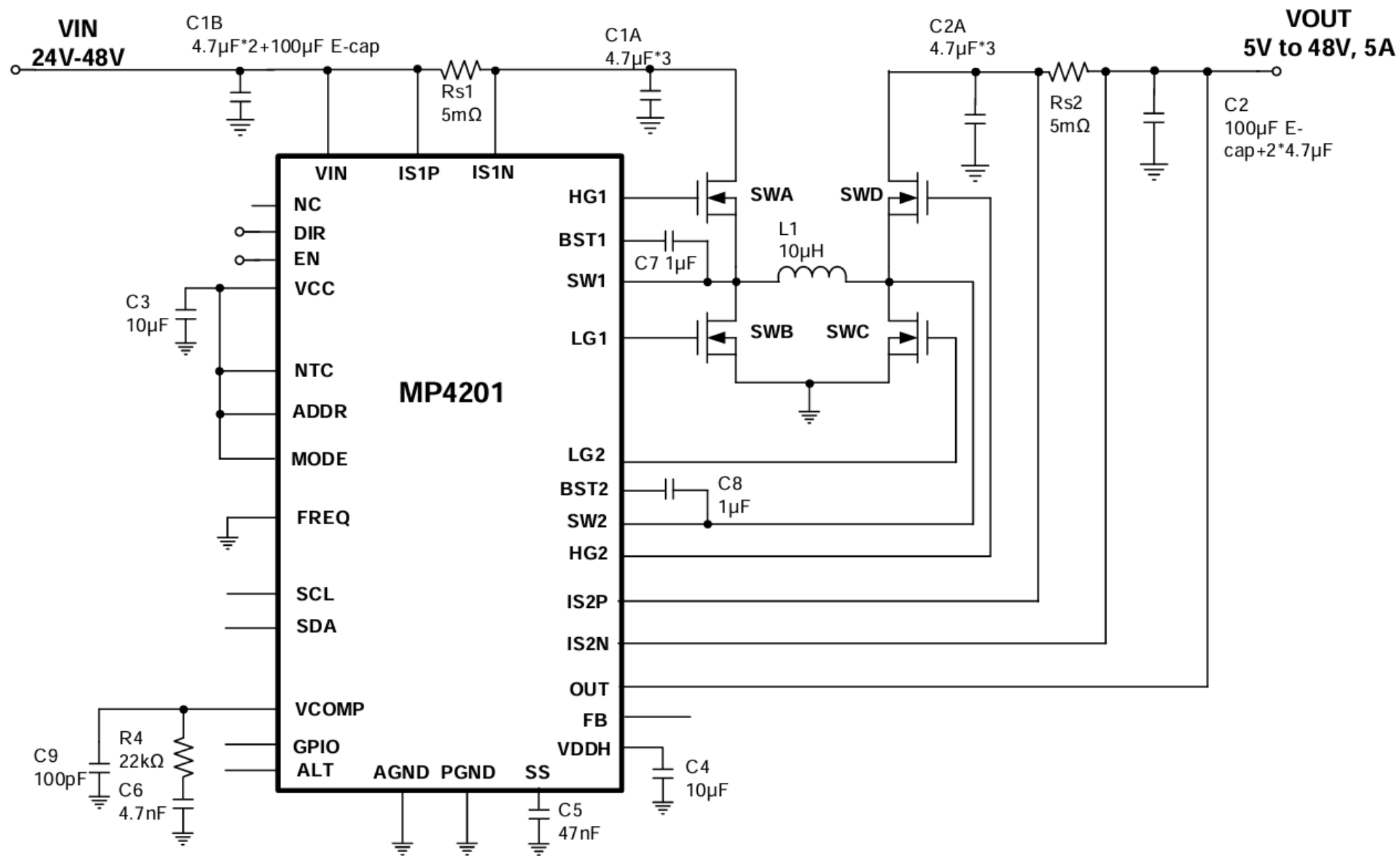
- USB EPR Power Delivery
- Generic High-Power Supply
- Solar Power (Battery charger, power Optimizer)
- Power Tool, Medical
- Powerbank

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Note:

- 1) Support input voltage <6V when bias external 7.7V~14V VDDH.

MP4201典型应用



仅供MP4201开源复刻
活动使用请勿外传

MP4201封装信息

ORDERING INFORMATION

Part Number*	Package	Top Marking	MSL Rating
MP4201GQJ-0000	QFN-32 (5mmx6mm)	See Below	1
MP4201GQJ-xxxx**			

* For Tape & Reel, add suffix -Z (e.g. MP4201GQJ-xxxx-Z).

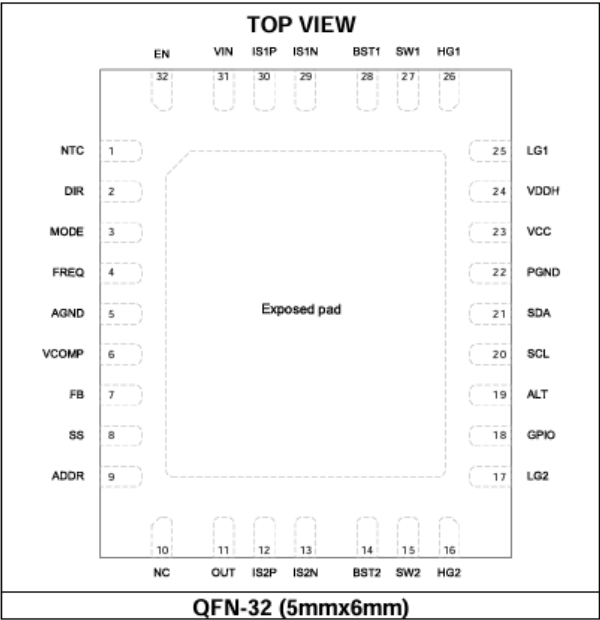
** "xxxx" is the configuration code identifier for the register settings stored in the OTP memory. Each "x" can be a hexadecimal value between 0 and F. MP4201-0000 is default code, and it can be programmed 1 time by user, other suffix code cannot be programmed by user any more.

TOP MARKING

MPSYYWW
MP4201
LLLLLLL

MPS: MPS prefix
YY: Year code
WW: Week code
MP4201: Part number
LLLLLLL: Lot number

PACKAGE REFERENCE



MP4201管脚定义

PIN FUNCTIONS

Pin # QFN-32 (5mm×6mm)	Name	Description										
1	NTC	The Temperature Sense Pin. It can be used for PCB board temperature sense and enter thermal shutdown when low voltage threshold is triggered. It should be connected to Vcc if not used.										
2	DIR	Buck-boost direction control. The DIR pin status and the register DIR control bit value will determine the buck-boost power flow direction.										
3	MODE	Auto PFM/PWM mode, forced PWM mode and frequency spread spectrum selection. <table><tr><td>MODE Pin voltage</td><td>Operation Mode</td></tr><tr><td>MODE=GND</td><td>PFM mode w/o FSS</td></tr><tr><td>MODE=1/3*VCC</td><td>PFM mode w/ FSS</td></tr><tr><td>MODE=2/3*VCC</td><td>FCCM mode w/o FSS</td></tr><tr><td>MODE=VCC</td><td>FCCM mode w/ FSS</td></tr></table>	MODE Pin voltage	Operation Mode	MODE=GND	PFM mode w/o FSS	MODE=1/3*VCC	PFM mode w/ FSS	MODE=2/3*VCC	FCCM mode w/o FSS	MODE=VCC	FCCM mode w/ FSS
MODE Pin voltage	Operation Mode											
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MODE=1/3*VCC	PFM mode w/ FSS											
MODE=2/3*VCC	FCCM mode w/o FSS											
MODE=VCC	FCCM mode w/ FSS											
4	FREQ	Frequency selection. <table><tr><td>FREQ Pin voltage</td><td>Operation frequency</td></tr><tr><td>FREQ =GND or float</td><td>200kHz</td></tr><tr><td>FREQ =1/3*VCC</td><td>400kHz</td></tr><tr><td>FREQ =2/3*VCC</td><td>600kHz</td></tr><tr><td>FREQ =VCC</td><td>1MHz</td></tr></table>	FREQ Pin voltage	Operation frequency	FREQ =GND or float	200kHz	FREQ =1/3*VCC	400kHz	FREQ =2/3*VCC	600kHz	FREQ =VCC	1MHz
FREQ Pin voltage	Operation frequency											
FREQ =GND or float	200kHz											
FREQ =1/3*VCC	400kHz											
FREQ =2/3*VCC	600kHz											
FREQ =VCC	1MHz											
5	AGND	Analog ground. Connect the AGND to PGND together.										
6	VCOMP	Internal error amplifier output pin. Connect a capacitor and a resistor in series to AGND for loop compensation.										
7	FB	Output feedback. Connect FB to the tap of an external resistor divider from the output to AGND to set the OUT voltage. If use the internal feedback setting by i2c register (0xD0[3] = 1b), FB pin can be float. Recommend to use internal feedback for bi-direction application.										
8	SS	Soft-start time setup. Connect a capacitor to SS pin.										
9	ADDR	The I2C slave address set pin. The chip's I2C slave address is determined by the ADDR pin status and register setup. This pin can also setup the default operation on/off, refer to the ADDR function description.										
10	N/C	No connection. It can be connected to GND or float.										
11	OUT	VOUT voltage sense input. Supplies power to VDDH based on VDDH power logic. Connect to the output capacitor. It is also used for the internal feedback divider. Route the OUT with wide PCB trace.										
12	IS2P	Positive terminal of average current limit sense input 2. Kelvin connect IS2P and IS2N to current sense resistor.										
13	IS2N	Negative terminal of average current limit sense input 2. Kelvin connect IS2P and IS2N to current sense resistor.										
14	BST2	Bootstrap power pin for boost high-side MOSFET gate driver. Connect one capacitor between BST2 and SW2. BST2 is supplied by VCC or BST1.										
15	SW2	Boost switch node of the converter. Connect to the source of SWD and the drain of SWC.										
16	HG2	Boost high-side MOSFET gate driver pin. Connect directly to the gate of SWD.										
17	LG2	Boost low-side MOSFET gate driver pin. Connect directly to the gate of SWC.										

MP4201管脚定义

PIN FUNCTIONS (continued)

Pin # QFN-32 (5mm×6mm)	Name	Description						
18	GPIO	Power good indication. When the output is good, this pin is open drain output; when the output is not good (OV, UV), this pin will be pulled low.						
		Battery charging status indication. Open drain structure, when battery charging is on-going, the GPIO is output PWM signal, 2s period, 50% duty cycle; when the charging is completed, this pin will be open drain output until reset operation bit (0x01[7] =0->1).						
		<table><tr><th>Battery Status</th><th>GPIO</th></tr><tr><td>Battery is in Charging</td><td>2s Period, 50% duty cycle.</td></tr><tr><td>Charge is completed</td><td>Open drain output</td></tr></table>	Battery Status	GPIO	Battery is in Charging	2s Period, 50% duty cycle.	Charge is completed	Open drain output
		Battery Status	GPIO					
Battery is in Charging	2s Period, 50% duty cycle.							
Charge is completed	Open drain output							
External clock synchronization input pin. The external clock synchronization frequency needs be close to the 90% to 110% of the FREQ pin frequency setup.								
19	ALT	Interrupt for different events in STATUS Word register. It is an open-drain output, and is pulled low when an interrupt event occurs, recovering to open drain when the fault is cleared. ALT is an open drain when the IC is not enabled.						
20	SCL	I²C clock signal.						
21	SDA	I²C data signal.						
22	PGND	Power ground. Gate-driving current return pin.						
23	VCC	Internal 5V LDO output. It is used for internal analog circuit supply. Decouple with a 10μF ceramic capacitor as close to this pin as possible. It is powered by VDDH.						
24	VDDH	Internal 7.2V LDO output. It is used for internal VCC bias supply. Powered by VIN or VOUT. Decouple with a 10μF ceramic capacitor as close to this pin as possible. It can be biased by external power supply (>7.7V) to improve efficiency.						
25	LG1	Buck low-side MOSFET gate driver pin. Connect directly to the gate of the SWB.						
26	HG1	Buck high-side MOSFET gate driver pin. Connect directly to the gate of SWA.						
27	SW1	Buck switch node of the converter. Connect to the source of SWA and the drain of SWB.						
28	BST1	Bootstrap power pin for buck high-side MOSFET gate driver. Supplied by VCC or BST2. Connect one capacitor between BST1 and SW1.						
29	IS1N	Negative terminal of average current limit sense input 1. Kelvin connect IS1P and IS1N to current sense resistor.						
30	IS1P	Positive terminal of average current limit sense input 1. Kelvin connect IS1P and IS1N to current sense resistor.						
31	VIN	VIN voltage sense input. Supplies power to VDDH based on VDDH power logic. Connect to the input capacitor. It is also used for the internal feedback divider. Route the VIN with wide PCB trace.						
32	EN	Enable control pin. Pull the EN pin to logic high to turn on the IC. Pull EN to logic low to turn off the IC. The EN pin has an internal 2MΩ pull-down resistor.						
Exposed Pad	-	Connect it to Ground.						

MP4201绝对最大额定值

ABSOLUTE MAXIMUM RATINGS ⁽²⁾

$V_{IN}, IS1P, IS1N, VOUT, IS2P, IS2N$	-0.3V to +105V
$V_{SW1}, SW2$	-1V to +105V (-5V to +108V for <20ns)
$V_{LG1}, LG2$	-0.3V to +6.5V (-2V to +8V for <20ns)
$V_{BST1}, HG1$	-0.3V to $V_{SW1} + 6.5V$
$V_{BST2}, HG2$	-0.3V to $V_{SW2} + 6.5V$
V_{EN}	-0.3V to 100V
V_{DDH}	-0.3V to +14V
All other pins	-0.3V to +6.5V

Junction temperature	150°C
Lead temperature	260°C
Storage temperature	-65°C to +150°C

ESD Rating ⁽⁴⁾

All pins (HBM)	± 2kV
All pins (CDM)	± 750V

Recommended Operating Conditions ⁽⁵⁾

Input voltage (V_{IN})	4.75V ⁽⁶⁾ to 100V
Output voltage (V_{OUT})	1V to 100V
Operating junction temp (T_J)	-40°C to +150°C

Thermal Resistance θ_{JA} θ_{JC_BOT}

EVQ4201 ⁽⁷⁾	TBD	TBD	°C/W
JESD51-7 ⁽⁸⁾	23.8	1.4	°C/W

ψ_{JT} ψ_{JB}

EVQ4201 ⁽⁷⁾	TBD	TBD	°C/W
JESD51-7 ⁽⁸⁾	0.3	4.5	°C/W

Notes:

- Exceeding these ratings may damage the device.
- The maximum allowable power dissipation is a function of the maximum junction temperature T_J (MAX), the junction-to-ambient thermal resistance θ_{JA} , and the ambient temperature T_A . The maximum allowable continuous power dissipation at any ambient temperature is calculated by P_D (MAX) = $(T_J$ (MAX) - T_A) / θ_{JA} . Exceeding the maximum allowable power dissipation will cause excessive die temperature, and the regulator will go into thermal shutdown. Internal thermal shutdown circuitry protects the device from permanent damage.
- HBM, per JEDEC specification JESD22-A114; CDM, per JEDEC specification JESD22-C101. JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process. JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.
- The device is not guaranteed to function outside of its operating conditions. When using internal feedback resistor, the voltage range is 3.2V to 81.92V; for other voltage setup, it needs to add external feedback resistor.
- Support input voltage <6V when bias external 7.7V~14V V_{DDH} .
- Measured on 6-layer PCB, 2oz-1oz-1oz-1oz-1oz-2oz. The junction to top characterization parameter, ϕ_{JT} , estimates the junction temperature in the real system, based on equation $T_J = \phi_{JT} \times P_{LOSS} + T_{CASE_TOP}$, where P_{LOSS} is the entire loss of IC at real application, T_{CASE_TOP} is case top temperature.
- The values given in this table are only valid for comparison with other packages and cannot be used for design purposes. These values were calculated in accordance with JESD51-2 and JESD51-14, and simulated on a specified JEDEC board. They do not represent the performance obtained in an actual application, the value of θ_{JC_BOT} shows the thermal resistance from junction-to-case bottom, θ_{JA} shows the thermal resistance from junction-to-ambient, and the value of ψ_{JT} shows the characterization parameter from junction-to-case top, ψ_{JB} shows Junction-to-board thermal characteristic parameter

MP4201电气特性

ELECTRICAL CHARACTERISTICS

$V_{IN} = 24V$ or $V_{OUT} = 24V$, $V_{EN} = 2V$, $T_J = -40^{\circ}C$ to $150^{\circ}C$, typical values are tested at $T_J = 25^{\circ}C$, unless otherwise noted.

Parameter	Symbol	Condition	Min	Typ	Max	Units
System						
Shutdown current	I_{Q_OFF}	$V_{EN} = 0V$, $T_J = 25^{\circ}C$			1	μA
		$T_J = -40$ to $+150^{\circ}C$			15	μA
Quiescent current	I_Q	Operation bit = 0, measured at both direction		300		μA
		Operation bit = 1, No switching, PFM mode. measured at both direction		2.7		mA
Input UVLO rising	$V_{IN_UVLO_R}$	Measure V_{IN} in charging mode, Measure V_{OUT} in discharging mode	4.2	4.45	4.7	V
Input UVLO hysteresis	$V_{IN_UVLO_HYS}$			350		mV
Operating VDDH voltage	V_{VDDH}	trim option	-5%	7.25	+5%	V
VDDH UVLO rising	$V_{DDH_UVLO_R}$		4.15	4.35	4.55	V
VDDH UVLO hysteresis	V_{DDH_HYS}			300		mV
VDDH power source change rising threshold	V_{SRC_TRANS}	V_{IN} or $V_{OUT} = 24V$, ramp V_{OUT} or V_{IN} from 5V to 24V. For default 7.2V voltage.	11.7	12.2	12.7	V
VDDH power source change hysteresis	$V_{SRC_TRANS_HYS}$			300		mV
Operating VCC voltage	V_{CC}	$V_{IN} = 24V$ or $V_{OUT} = 24V$, 0mA.	-2%	5.2	+2%	V
		$V_{IN} = 24V$ or $V_{OUT} = 24V$, 50mA Load.	-5%	5.2	+5%	V
VCC UVLO rising voltage	$V_{CC_UVLO_R}$		4.15	4.3	4.45	V
VCC UVLO hysteresis	V_{DDH_HYS}			500		mV
Thermal shutdown ⁽⁹⁾	T_{TSD}		155	165	175	$^{\circ}C$
Thermal shutdown hysteresis ⁽⁹⁾	T_{TSD_H}			20		$^{\circ}C$
EN rising to enable VCC threshold	$V_{EN_R_VCC}$		0.45	0.75	1.05	V
EN rising to enable switching threshold voltage	$V_{EN_R_SW}$		1.3	1.4	1.5	V
EN voltage to enable switching hysteresis voltage	V_{EN_H}			200		mV
EN pull-down resistance	R_{EN}	EN = 2V	1.6	2	2.4	M Ω
Buck-Boost Controller						
FB reference voltage (external feedback)	V_{REF}	$V_{OUT} = 80V$. $T_J = 0$ to $85^{\circ}C$	-1%	2.5	+1%	V
		$V_{OUT} = 48V$. $T_J = 0$ to $85^{\circ}C$	-1%	1.5	+1%	V
		$V_{OUT} = 5V$. $T_J = 0$ to $85^{\circ}C$	-2%	0.156 25	+2%	V

MP4201电气特性

Parameter	Symbol	Condition	Min	Typ	Max	Units
Vout voltage (internal feedback)	V _{BATT_REG}	T _J = 25°C	-1%	80	+1%	V
		T _J = 25°C	-1%	48	+1%	V
		T _J = 25°C	-1%	36	+1%	V
V _{REF} change slew rate rising	T _{REF_R}	I2C set	0.11	0.15	0.17	mV/μs
V _{REF} change slew rate falling	T _{REF_F}	I2C set	0.08	0.1	0.12	mV/μs
Output discharge current	I _{DISCHG}	I2C Set = 01b		50		mA
Buck-boost to buck ratio	R _{BB_BUCK}	VIN/VOUT, f _{SW} = 400kHz		120%		
Buck to buck-boost ratio	R _{BUCK_BB}	VIN/VOUT, f _{SW} = 400kHz		115%		
Buck-boost to boost ratio	R _{BB_BOOST}	VIN/VOUT, f _{SW} = 400kHz		80%		
Boost to buck-boost ratio	R _{BOOST_BB}	VIN/VOUT, f _{SW} = 400kHz		85%		
Current Limit						
Buck valley current limit	I _{LIMIT_VALLEY}	Switching current limit = 011b, Rsens = 5mΩ		22		A
Boost peak current limit	I _{LIMIT_PEAK}			25		A
Output UVP threshold	V _{UVP}	Measure OUT in charging mode, Measure VIN in discharging mode	35%	40%	45%	V _{REF}
Output enter hiccup threshold	V _{TH_HICCUP}	only works in no battery charging function	-5%	2.51	+5%	V
Average constant current limit for IS1P and IS1N	I _{LIMIT_1}	Current sense resistor × Current Limit	4.25	5	5.75	mV
			-5%	15	+5%	mV
			-5%	50	+5%	mV
Average constant current limit for IS2P and IS2N	I _{LIMIT_2}	Current sense resistor × Current Limit	-5%	15	+5%	mV
			-5%	20	+5%	mV
			-5%	50	+5%	mV
Negative current limit	I _{NOCP}			-5	-3	A
Switching Frequency						
Switching frequency	f _{SW}	FREQ = GND	-15%	200	+15%	kHz
		FREQ = 1/3×VCC	-15%	400	+15%	kHz
		FREQ = 2/3×VCC	-15%	600	+15%	kHz
		FREQ = VCC	-15%	1000	+15%	kHz
Frequency spread spectrum span	f _{SS}		±6%	±8%	±10%	F _{SW}
Frequency spread spectrum modulation frequency	F _{MODULATION}		-10%	2	+10%	kHz

MP4201电气特性

Parameter	Symbol	Condition	Min	Typ	Max	Units
Absolute output OVP hysteresis voltage	V _{OUT_ABS_HYS}		1	1.6	2	V
Input OVP Protection (Measure VOUT pin in charging mode, measured VIN in discharging mode)						
Absolute input OVP rising threshold 1	V _{IN_ABS_R1}		-3%	30.4	+3%	V
Absolute input OVP rising threshold 2	V _{IN_ABS_R2}		-3%	60.8	+3%	V
Absolute input OVP rising threshold 3	V _{IN_ABS_R3}		-3%	104	+3%	V
Absolute input OVP hysteresis voltage	V _{IN_ABS_HYS}		1	1.5	2	V

DIR input logic high voltage	V _{DIR_H}		2.5			V
DIR input logic low voltage	V _{DIR_L}				0.4	V
DIR pull-down resistor	R _{DIR}		1.6	2	2.4	MΩ
FREQ, MODE, ADDR						
The voltage range 1	V _{RANGE1}				0.15	VCC
The voltage range 2	V _{RANGE2}		0.25		0.45	VCC
The voltage range 3	V _{RANGE3}		0.55		0.75	VCC
The voltage range 4	V _{RANGE4}		0.85			VCC
Pull-down resistance	R _{INTERNAL}		1	2	3	MΩ
I2C slave address	ADDR	ADDR = GND		27h		
		ADDR = 1/3×VCC		2Fh		
		ADDR = 2/3×VCC		37h		
		ADDR = VCC		3Fh		
Charging status indication period time ⁽⁹⁾	T _{CHG_STATUS}		-10%	2	+10%	s

MP4201电气特性

I ² C Interface						
Input logic high	V _{IH}		1.35			V
Input logic low	V _{IL}				0.9	V
Output voltage logic low	V _{OUT_L}	Sink = 2mA			0.4	V
SCL clock frequency	f _{SCL}			400	1000	kHz
SCL high time ⁽⁹⁾	t _{HIGH}		260			ns
SCL low time ⁽⁹⁾	t _{LOW}		500			ns
Data setup time ⁽⁹⁾	t _{SU.DAT}		50			ns
Data hold time ⁽⁹⁾	t _{HD.DAT}		0	60		ns

Parameter	Symbol	Condition	Min	Typ	Max	Units
Setup time for (repeated) start condition ⁽⁹⁾	t _{SU.STA}		260			ns
Hold time for (repeated) start condition ⁽⁹⁾	t _{HD.STA}		260			ns
Bus free time between a start and a stop condition ⁽⁹⁾	t _{BUF}		500			ns
Setup time for stop condition ⁽⁹⁾	T _{SU.STO}		260			ns
Rise time of SCL and SDA ⁽⁹⁾	t _R		10		120	ns
Fall time of SCL and SDA ⁽⁹⁾	t _F		10		120	ns
Pulse width of suppressed Spike ⁽⁹⁾	t _{SP}		0		50	ns
Capacitance for each bus line ⁽⁹⁾	C _B				400	pF

Note:

9) All minimum and maximum parameters are tested at T_J = 25°C. Over-temperature limits are guaranteed by characterization and correlation.

MP4201功能框图

FUNCTIONAL BLOCK DIAGRAM

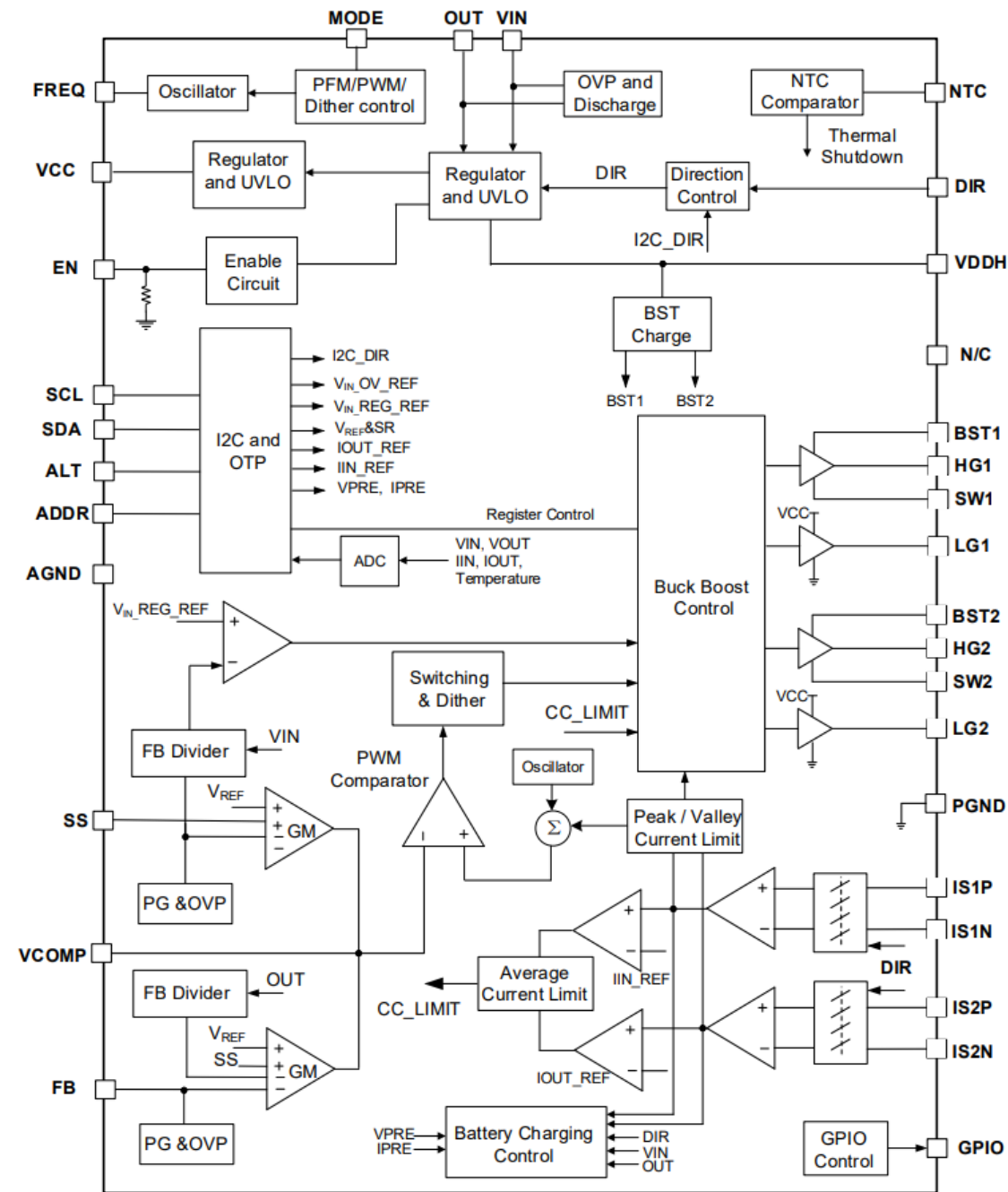


Figure 2: Functional Block Diagram

MP4201功能原理

OPERATION

The MP4201 integrates a bi-directional four-switch buck-boost controller. The buck-boost controller works with fixed frequency in buck, boost, and buck-boost modes. One special buck-boost control strategy provides high efficiency over the device's full input range and smooth transient between different modes.

Figure 2 shows the internal block diagram, and the following sections describe the MP4201's functions.

VCC Power Supply

The MP4201's internal circuit and four MOSFET gate driver are powered by VCC. VDDH is regulated from V_{IN} , V_{OUT} , while the VCC is regulated from the VDDH.

When V_{IN} power or V_{OUT} power is supplied and EN pin is high, the MP4201 tries to regulate VDDH and Vcc.

After the VCC rises above the UVLO threshold voltage, it will check the direction firstly; then MP4201 will check if the input voltage is higher than the programmable input UVLO threshold. If the input voltage is higher than the Input UVLO SEL threshold, the part starts switching if operation is enabled, and regulates output by soft-start control.

If V_{IN} and V_{OUT} are both above 12.2V, the MP4201 powers VDDH from whichever is the lower voltage source to reduce power-loss. Otherwise, it powers VDDH from the higher voltage power source of V_{IN} and V_{OUT} to get enough VDDH voltage.

Table 1: VDDH Voltage Supply

V_{IN} , V_{OUT}	$V_{IN} > 12.2V$ and $V_{OUT} > 12.2V$	$V_{IN} < 12.2V$ or $V_{OUT} < 12.2V$
VDDH Supply	Which one is lower in [V_{IN} , V_{OUT}]	Which one is higher in [V_{IN} , V_{OUT}]

VDDH and BST have separate UVLO, which keeps the gate signal off when the voltage is low. VDDH and BST should have enough voltage to enable MP4201 switching.

Enable Control (EN)

The EN pin enables and disables the MP4201. When applying a voltage higher than the EN high threshold, the MP4201 enables to start switching if operation is default on. Switching operation is disabled and VCC voltage will be shut down when the EN voltage falls below its lower threshold.

For bi-directional operation, suggest to connect two oring diodes from V_{IN} and V_{OUT} to control the EN. Add a divider resistors for start-up timing control. After EN shutdown, the I²C resets to its default value after a 300ms discharge time.

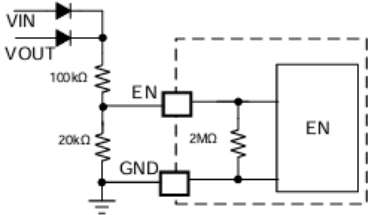


Figure 3: EN Connection in Bi-Directional Operation

Programmable Input Under-Voltage Lockout (UVLO)

Under-voltage lockout (UVLO) protects the chip from operating at an insufficient supply voltage. The MP4201 is enabled when VCC voltage is higher than its rising UVLO threshold.

The MP4201 input voltage UVLO can be programmed by I²C register setup. When set in charging mode, the buck-boost starts switching after V_{IN} pin voltage is higher than the input UVLO threshold; the MP4201 stops switching after V_{IN} pin voltage is below its UVLO falling threshold.

When set in discharging mode, it monitors the OUT pin for programmable UVLO. Suggest to set the minimum input UVLO threshold at battery side, the buck-boost starts switching after OUT pin voltage is higher than the minimum input UVLO threshold; the MP4201 stops switching after OUT pin voltage is below the input UVLO falling threshold.

When the MP4201 is powered off by the VCC UVLO or the EN pin signal, the I²C interface cannot respond to the host, soft-start voltage is

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discharged. The BST voltages drop slowly with leakage, but all logic is off.

Thermal Protection

The MP4201 integrates one temperature monitor circuit. If the junction temperature is higher than OTP threshold (typically 165°C), the MP4201 shuts down; after the temperature drops below 140°C, the IC resumes operation. If the junction temperature is higher than the over temperature warning threshold (OTW), the MP4201 doesn't shutdown but report the event to STATUS_TEMPERATURE register.

When OTP or OTW is triggered, ALT pin is pulled low if it is not masked. The OTP and OTW threshold can be programmed by I2C register.

Buck-Boost Operation

The MP4201 can regulate output above, equal to, or below the input voltage. Based on the one-inductor, four-switch power structure (see Figure 4), it operates in buck mode, boost mode, or buck-boost mode with different V_{IN} inputs (see Figure 5).

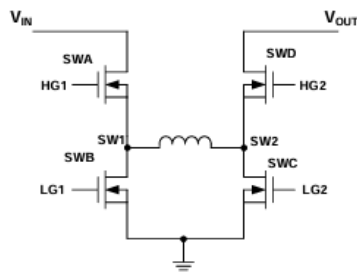


Figure 4: Buck-Boost Topology

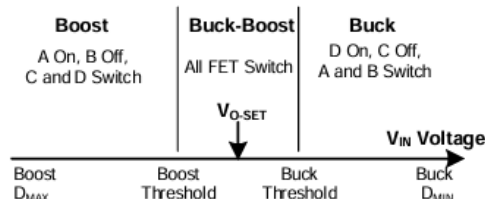


Figure 5: Buck-Boost Operation Range

Buck Mode ($V_{IN} > V_{OUT}$)

When V_{IN} exceeds V_{OUT} , the MP4201 works in buck mode. In buck mode, switch A (SWA) and switch B (SWB) switch for buck regulation,

switch C (SWC) is off, and switch D (SWD) remains on to conduct the inductor current.

In each cycle, SWA turns on to conduct the inductor current. SWA turns off until the inductor current reaches the COMP voltage (VCOMP), and SWB turns on until the tCLK times out. An internal oscillator controls SWA's off time to maintain a fixed frequency. The COMP signal is an error amplifier (EA) output from the VOUT feedback and internal feedback reference voltage (VFB). Therefore, VCOMP controls the inductor current to supply the output terminal voltage. Figure 6 shows the buck mode waveform.

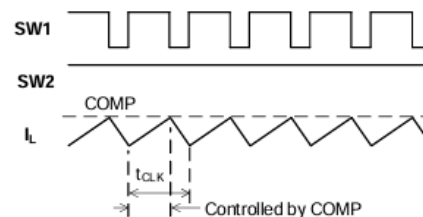


Figure 6: Buck Mode Waveform

Boost Mode ($V_{IN} < V_{OUT}$)

When V_{IN} is below V_{OUT} , the MP4201 works in boost mode. In boost mode, SWC and SWD switch for boost regulation, SWB is off, and SWA remains on to conduct the inductor current.

In each cycle, SWC turns on to conduct the inductor current. When the inductor current rises and triggers the control signal on the COMP pin, SWC turns off until tCLK times out to boost I_L to the output. While SWC is off, SWD turns on for the current freewheeling period. An internal oscillator controls the SWC off time to maintain a fixed frequency. Then, SWC turns on and off repeatedly to regulate the current to match the COMP signal. V_{COMP} controls the inductor current to supply the output terminal voltage in boost mode. Figure 7 shows the boost mode waveform.

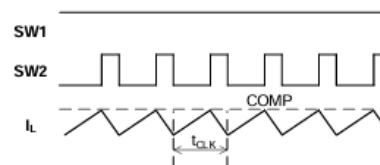


Figure 7: Boost Mode Waveform

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Buck-Boost Mode ($V_{IN} \approx V_{OUT}$)

When V_{IN} is almost equal to V_{OUT} , the converter cannot provide enough energy to the load in buck mode due to SWA's minimum off time. In boost mode, the converter supplies excess power to the load due to SWC's minimum on time. Under these conditions, the MP4201 adopts buck-boost control to regulate the output (see Figure 8).

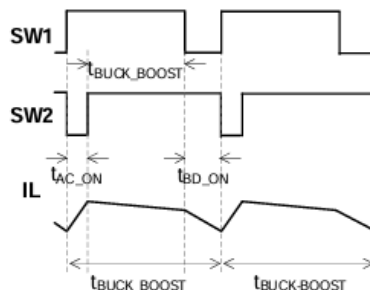


Figure 8: Buck-Boost Mode Waveform

Buck-boost mode is activated if V_{IN} is almost equal to V_{OUT} . The MOSFET turn-on sequence is as follows:

1. SWA and SWC
2. SWA and SWD
3. SWB and SWD

Via this process, the inductor current reaches the V_{COMP} requirement and supplies enough current to the output.

If the buck mode minimum off time is almost reached, the IC enters buck-boost mode. For $f_{SW} = 400\text{kHz}$, If V_{IN} is about 20% above V_{OUT} in buck-boost mode, the IC changes to buck mode. If V_{IN} is about 20% below V_{OUT} , the IC enters boost mode. If the boost mode minimum on time is almost reached, the IC enters buck-boost mode. For $f_{SW} = 200\text{kHz}$, the threshold changes to 12.5%.

Forced CCM Mode

The MP4201 supports forced continuous conduction mode (FCCM) operation. When the load decreases, the average input current drops and the inductor current may go to negative during the SWD on-period. This forces the inductor current to work in continuous mode with fixed frequency, producing a low V_{OUT} ripple.

PFM (Auto PFM/PWM Mode)

In PFM condition, once the inductor current drops to ZCD threshold, SWD turns off to prevent the current from flowing from V_{OUT} to GND in discharging mode, forcing the inductor current to work in discontinuous conduction mode (DCM). Simultaneously, the internal off-time clock stretches once the MP4201 enters DCM mode. The frequency drops when the inductor current conduction period decreases, helping to save power loss and reduce the V_{OUT} ripple.

If V_{COMP} drops to the PFM threshold, the MP4201 stops switching to decrease more switching power loss. The MP4201 recovers switching once V_{COMP} rises above the PFM threshold. The switching pulse skips based on V_{COMP} in very light-load condition. PFM has a much higher efficiency than FCCM mode in light load, but the V_{OUT} ripple may be higher due to the group switching pulse.

Input and Output Constant Current Limit

The MP4201 has both input and output current limit control. A sense resistor can be connected to the VIN and VOUT line for average output current limit control.

When set in charging mode, the power flow is from VIN to OUT, the IS1P and IS1N pins are used for input constant current limit setup; IS2P/IS2N pins are used for output current limit setup. When set in discharging mode, the power flow is from OUT to VIN, the IS1P and IS1N pins are used for output constant current limit setup; IS2P/IS2N pins are used for input current limit setup.

When output overload occurs, the MP4201 limits the output current by average current limit loop regulation. The output voltage decrease until hiccup mode (OCP_MODE=0) is triggered or stay in constant current mode (OCP_MODE=1).

When the input average current limit is triggered, the MP4201 will limit the input current by average current limit loop regulation and the output voltage decrease until hiccup mode (OCP_MODE=0) is triggered or stay in constant current mode (OCP_MODE=1).

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The OCP_MODE control is active for both directions. The input and output average current limit can be enabled/disabled by I2C register setup.

Switching Current Limit

The MP4201 senses SWA and SWD current in the loop control, it provides valley current limit in buck mode and peak current limit in boost mode in each cycle-by-cycle switching. In buck mode, the next period will not start before I_L drops to the valley current limit, so it may fold-back the frequency when the valley current limit is triggered. The switching current limit can be programmed with an external sense resistor and switching current limit register setup.

After input average current limit, output average current or switching current limit is triggered, the FB voltage drops below 40% of V_{REF} and V_{out} is $<2.51V$, it will enter hiccup mode when set OCP_MODE=0. This protection mode is especially useful when the output is dead-short to ground. This greatly reduces the average short circuit current, alleviates thermal issues, and protects the regulator. The MP4201 exits hiccup mode once the over current condition is removed. The hiccup function only works in no charging mode. The hiccup is automatically disabled in charging mode.

If the OCP_MODE=1, hiccup is disabled, the IC continues switching with average current limit setting or cycle-by-cycle current limit for both directions.

Output Voltage Regulation

The MP4201 can regulate the output voltage through internal feedback or external feedback for both directions.

When set in charging mode, the power supply terminal is at VIN pin, the regulation voltage terminal is at the OUT pin. The MP4201 regulates output voltage through FB pin for external feedback or internal feedback through OUT pin. The regulation voltage is controlled by VOUT_COMMAND register value, the reference voltage range is from 0.1V to 2.56V.

When set in discharging function, the power supply terminal is at OUT pin, the regulation voltage terminal is at the VIN pin. The MP4201 regulates the output voltage through internal feedback at VIN pin or FB pin with external

feedback setup. The regulation voltage is still controlled by VOUT_COMMAND register value, the reference voltage range is from 0.1V to 2.56V.

The Minimum Input Operation Voltage Regulation (VIN_REG)

The MP4201 can also set the minimum input voltage operation threshold by changing VIN_REG_THLD register value. Input regulation function can work in charging mode, it monitors the VIN pin voltage. In discharging mode, suggest to disable the input regulation function at battery side.

When the output load current exceeds the power supply capability result in power supply input voltage drops to trigger the programmed VIN_REG_THLD threshold, the MP4201 will decrease the output voltage to maintain the input voltage; In the CCM mode, if the input voltage still drops to lower than VIN_REG_THLD threshold, and output current is $<500mA$, deglitch timer (2ms, typical) the switching will be disabled and enter hiccup, which is to stop converting power from input to output. After hiccup timer is expired, the switching will be enabled. In the PFM mode, if the input voltage still drops to lower than VIN_REG_THLD threshold, the switching will be disabled until the output current is reduced to 0.

Switching Frequency and Frequency Spread Spectrum Function

The MP4201 programs switching frequency with with FREQ pin. The frequency is selectable at 200kHz, 400kHz, 600kHz, 1MHz.

Table 2: FREQ Pin Setup

FREQ Pin voltage	Operation frequency
FREQ =GND or float	200kHz
FREQ =1/3*VCC	400kHz
FREQ =2/3*VCC	600kHz
FREQ =VCC	1MHz

The MP4201 has a frequency spread spectrum (FSS) function. Set dither enable=1 (MODE pin =VCC or 1/3Vcc) can enable the FSS function; set dither enable=0 (MODE pin=GND or 2/3Vcc) can disable the FSS function. The purpose of the spread spectrum is to minimize the peak emissions at certain frequencies.

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Typically, the MP4201 uses a 2kHz triangle wave to modulate the internal oscillator. The frequency span of the spread spectrum operation is $\pm 8\%$.

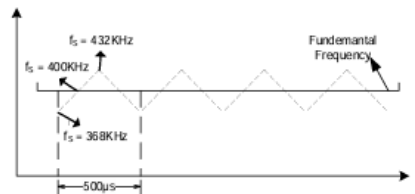


Figure 9: Frequency Spread Spectrum

The MP4201 frequency spread frequency can be enabled/disabled for 200kHz, 420kHz, 600kHz, 1MHz switching frequency.

When GPIO is programmed as frequency synchronization function, the switching frequency SYNC range is from 90% to 110% of the default FREQ pin setup value.

Suggest to pull low EN pin or set OPERATION = 0 firstly, then change the FREQ pin status.

Gate Driver and BST Power

The MP4201 provides four N-channel MOSFET gate drivers for the H-bridge MOSFETs (see Figure 4). Each driver is capable of sourcing and sinking current. In buck operation, LG1 and HG1 switch while HG2 remains on. In boost operation, LG2 and HG2 switch while HG1 remains on. LG1 and LG2 are powered by VCC power, while HG1 and HG2 are powered by BST1 and BST2 power.

Capacitors between BST1 to SW1 and BST2 to SW2 are necessary to supply the power, which can be from an internal diode from VCC or from charging each other.

The BST power has its own UVLO control.

Soft Start (SS)

To avoid overshoot during start-up, the MP4201 has built-in soft start (SS) to ramp up V_{OUT} with a controlled slew rate, connect a capacitor to SS pin. The SS charge current is about $6\mu\text{A}$. the soft-start time can be calculated as below:

$$T_{SS} (\text{ms}) = V_{REF} (\text{V}) \times C_{SS} (\text{nF}) / I_{SS} (\mu\text{A}) \quad (1)$$

Typically, the I_{SS} is $6\mu\text{A}$.

When the output is shorted to GND, the feedback voltage is pulled low, and V_{SS} is

discharged. The part initiates a normal soft start when it returns to a normal state.

Output Over-Voltage Protection

When set in charging mode, the MP4201 monitors FB pin voltage (external feedback) or OUT pin voltage (internal feedback); When set in discharging mode, the MP4201 monitors VIN pin voltage for output OVP protection.

If the output voltage exceeds 120% of V_{REF} , the IC stops switching and discharges the V_{OUT} capacitor voltage through one internal discharge resistor. It stops discharging when output voltage drops to 110% of the regulation voltage. The function can be controlled by VOUT OVP_EN bit, 0xD6 register.

The MP4201 also integrates the absolute output over voltage protection. The absolute output over voltage enable control and threshold selection are programmed by ABS_VOUT_OVP register value. When absolute output over voltage threshold is triggered, the switching will be disabled and discharge path will be turned on.

There is a special OVP threshold control in battery charging mode (Battery charging EN=1). 104% battery OVP rising threshold, 102% battery OVP falling threshold.

Input Over-voltage Protection

The MP4201 supports input power supply over voltage protection. When power flow is from VIN pin to OUT pin, the MP4201 monitors VIN pin voltage for the input OVP protection; when power flow is from OUT pin to VIN pin, the MP4201 monitors the OUT pin voltage for the input OVP protection. The input OVP protection can be enabled/disabled by the I2C register setup, 0xD6.

When input over voltage rising threshold is triggered, it will stop switching and the ALT pin will be pulled low, output discharge path will be turned on. After the input voltage drops below the falling threshold, the MP4201 back to switching again.

The input over-voltage threshold can be adjusted by I2C register.

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Output Discharge

If the Battery Charge EN=1, there is no discharge function at the OUT pin in charging mode; in discharge mode, the output discharge is at the VIN pin.

If the Battery Charge EN=0, the MP4201 integrates output discharge function for both directions. The discharge function can be enabled/disabled by Discharge EN bit in I2C register.

The output discharge function is enabled in the following conditions:

1. The output OVP threshold (+120% of V_{FB} voltage) or absolute OVP is triggered; the discharge will be disabled after VOUT OVP is recovery.
2. I2C OPERATION bit off or EN pin off. Discharge works until 300ms delay time expired.
3. If input UVLO is triggered, but VCC has residual voltage, the MP4201 discharges in a limited time. This discharge function will be disabled after the VCC voltage drops below a certain voltage.
4. Output is changed from high to low, output discharge automatically.
5. Input OVP status is triggered.

Line Drop Compensation

The MP4201 is capable of compensating the voltage drop, such as high impedance caused by a long trace, to keep a fairly constant load-side voltage. When set in charging mode, the line drop compensation is disabled automatically; when set in discharging mode, the line drop compensation is active for VIN pin. Refer to register description for detail line drop compensation amplitude.

The line drop compensation can be programmed by I2C register.

NTC Function

The MP4201 integrates a NTC pin that can be used to monitor the PCB board, MOSFET temperature by measuring the voltage ratio between the NTC pin's voltage and the VCC pin's voltage.

The MP4201 will shut down after the NTC voltage drops below the threshold; after the NTC

pin voltage back to normal, the MP4201 will back to switch again.

ADDR Function

The ADDR pin can program I2C slave address, meanwhile set the default I2C_EN status. The I2C_EN is the logic OR result of OPERATION bit value in I2C register (0x01) and ADDR pin status. Resistor divider on ADDR pin determines the A5:A4 bits values of 7-bit I2C address, and A3:A1 are I2C/OTP programmable.

Table 3: ADDR Pin Setup

I2C Address A7:A1, 01xxxxx			Default I2C_EN
GPIO_ADDR Status	Binary	Hex	
ADDR=GND	0100 111	27H	OFF
ADDR=1/3 ×VCC	0101 111	2FH	OFF
ADDR=2/3 ×VCC	0110 111	37H	ON
ADDR=VCC	0111 111	3FH	ON

Bi-Directional Operation

The MP4201 supports bi-directional buck-boost operation, and it can be used for battery pack charging mode and general buck-boost mode. Both DIR pin and DIR bit can control the power flow direction.

The MP4201 integrates three working modes:

- Battery Charging mode (VIN to OUT)
- Buck-boost mode (VIN to OUT) without charging function
- Buck-boost mode (OUT to VIN) or discharging mode

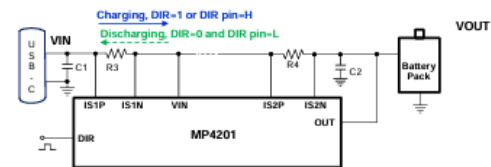


Figure 11: Bi-directional Buck-boost working

When set Battery Charge EN =1, DIR bit =1 or DIR pin=high, the MP4201 works in battery charging mode, the current flow is from VIN to OUT. The MP4201 supports up to 19s Li-ion battery pack charging. The charging current, battery regulation voltage can be programmed by I2C register. During charging mode, the output OVP is 104% of V_{OUT} rising with 2% hysteresis voltage; output discharge function,

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line drop compensation and OCP hiccup are disabled automatically.

When set DIR bit =0 and DIR pin=low, no matter the Battery Charging EN control bit value, the MP4201 works in buck-boost operation only, the current flow is from VOUT to VIN. This mode can be used for battery discharge function. If it is in battery discharge mode, the input UVLO, input OVP should be set at proper value that is determined by customer application, and suggest to disable input regulation function by I2C control as well.

When set Battery Charge EN =0, DIR bit =1 or DIR pin=high, the MP4201 works in buck-boost mode only, the current flow is from VIN to OUT, internal battery charging function doesn't work,

the output OVP is 120% of Vout, OCP mode is controlled by register setup.

If the power direction need be changed, the EN pin should be pulled low or set I2C EN=0 (0x01=0), then DIR logic can be changed; when the EN pin=1 or I2C EN=1, the buck-boost direction status is latched and cannot be changed again.

When output discharge function is disabled, there is no 300ms output shutdown discharge time, so the system can be reset after inductor current drops to zero, it is OK to change the direction again; when output discharge function is enabled, it needs extra 300ms discharge time to reset the system, so suggest to add 500ms delay time to change the direction again.

Table 4: Bi-direction function ⁽¹⁰⁾

Direction	Working Mode	Conditions	Input UVLO	Input OVP	Input Regulation	Input Current Limit	Output OVP	Output Current limit	Output Discharge	Line drop compensation
Charge	Charger mode (VIN to OUT)	Battery Charging EN=1, DIR pin=high or DIR bit=1	Yes. Monitor VIN pin. Programmable UVLO.	Yes. Monitor VIN pin.	Yes. Monitor VIN pin.	Yes. Monitor IS1P-IS1N.	No ABS OVP. FB OVP threshold is 104% rising, 2% hysteresis.	Yes. Monitor IS2P-IS2N, Hiccup is disabled.	No	No
	Buck-boost mode (VIN to OUT) ⁽¹⁰⁾	Battery Charging EN=0, DIR pin=1 or DIR bit=1	Yes. Monitor VIN pin. Programmable UVLO.	Yes. Monitor VIN pin.	Yes. Monitor VIN pin.	Yes. Monitor IS1P-IS1N.	Yes. Both ABS OVP (monitors OUT pin) and FB OVP work. ⁽¹⁰⁾	Yes. Monitor IS2P-IS2N. Hiccup is controlled by register setup.	Yes. It is active on OUT pin.	Yes. It is active on OUT pin.
Discharge	Buck-boost mode (OUT to VIN)	Battery Charging EN=X, DIR pin=0 and DIR bit=0	Yes. Monitor OUT pin. Programmable UVLO.	Yes. Monitor OUT pin.	Yes. Monitor OUT pin.	Yes. Monitor IS2N-IS2P.	Yes. Both ABS OVP (monitors VIN pin) and FB OVP work. ⁽¹⁰⁾	Yes. Monitor IS1N-IS1P. Hiccup is controlled by register setup.	Yes. It is active on VIN pin.	Yes. It is active on VIN pin.

Note:

¹⁰⁾ When set in battery discharge function (Buck-boost mode, OUT to VIN), suggest to set proper input UVLO threshold, disable input OVP and input regulation function.

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Charge Cycle Control

The MP4201 integrates battery charging function: Pre-charge, constant current, constant voltage, termination charging detection and auto re-charge control.

When the control bit Battery Charge EN =1 and OPERATION=1, the charging cycle is enabled. MP4201 starts to check the battery voltage firstly, and then determine which charging cycle is triggered, the OCP hiccup mode is disabled no matter OCP_MODE setup, and also output OVP threshold is changed to 104% rising, 102% falling of VOUT_COMMAND voltage without discharge function. When the battery voltage exceeds the OVP threshold, the battery charging will be disabled; after the battery voltage is below than the OVP threshold, the battery charging will be enabled again.

When the Battery Charge EN =0, the charging function is disabled, the MP4201 works as a general buck-boost controller function with CC/CV function, the OCP_MODE setup will determine it enters hiccup or not, and output OVP is controlled by the resister setup.

When choose the pre-charge or termination current <400mA, the switching frequency is suggested 400kHz to improve the current sense accuracy.

Phase 1 Pre-charge: if the battery voltage is below than the Pre-Charge Threshold ($V_{BATT} < V_{PRE}$), the MP4201 will enter pre-charge status. The charge current is determined by the Pre Charge Current register setup. The pre-charge can be disabled by I2C register.

Phase 2 Constant current fast charge: when the battery voltage is higher than Pre-charge Threshold setup ($V_{BATT} > V_{PRE}$), the MP4201 will enter constant current fast charge status. The charge current is determined by the OUT_CURRENT_LIMIT register setup.

Phase 3 Constant voltage charge: when the battery voltage reaches the battery regulation voltage, the charge current will decrease until the termination current threshold is triggered.

After the termination charge current is triggered ($I_{CHG} < I_{TERM}$) with the deglitch timer expired (typical 500ms), the battery charge is completed, and the MP4201 will report it to upper MCU controller by pulling low ALT pin and set the register STATUS_WORD, D[9]=1. At the same time, if the EN_TERM_CHG=1, the charging will be disabled unless new charge cycle is enabled. If the EN_TERM_CHG=0, the charging will not be disabled unless the operation is off.

Auto Recharge

After the battery charging is terminated by set EN_TERM_CHG=1, the charge will be disabled. The battery voltage may be decreased due to external load or self- discharge. When the battery voltage is below than the recharge threshold ($V_{BATT} < V_{RE_CHG}$ is 95% of the VOUT_COMMAND value) with deglitch timer is expired (typical 500ms), new charging cycle will be enabled if the VIN and EN are present, Battery Charge EN=1.

A new charging cycle is reset by following event:

1. OPERATION status is changed;
2. VIN or VCC UVLO or EN UVLO is triggered.
3. Recharge Cycle is triggered

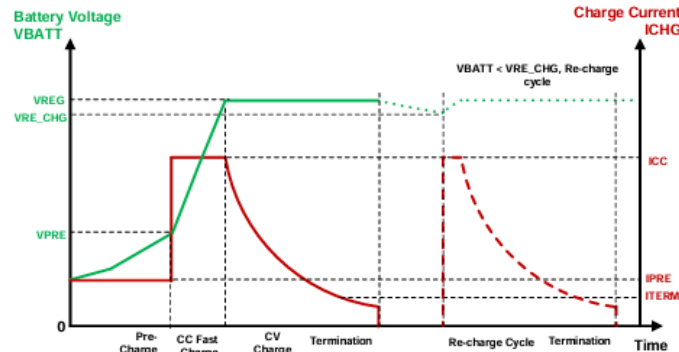


Figure 12: Charge Cycle Control

MP4201寄存器配置

PMBUS INTERFACE

PMBus Serial Interface Description

The power management bus (PMBus) is an open-standard power management protocol that defines a means of communication with power conversion and other devices.

The PMBus is a two-wire, bidirectional serial interface, consisting of a data line (SDA) and a clock line (SCL). The lines are pulled to a bus voltage externally when they are idle. When connecting to the lines, a master device generates an SCL signal and device address, and arranges the communication sequence. This is based on I²C operation principles.

Start and Stop Conditions

The start and stop conditions are signaled by the master device, which signifies the beginning and end of the PMBus transfer. The start condition is defined as the SDA signal transitioning from high to low while the SCL is high. The stop condition is defined as the SDA signal transitioning from low to high while the SCL is high (see Figure 13).

The master then generates the SCL clocks and transmits the device address and the read/write direction bit (R/W) on the SDA line. Data is transferred in 8-bit bytes by the SDA line. Each byte of data is followed by an acknowledge bit (ACK).

PMBus Update Sequence

The MP4201 requires a start condition, a valid PMBus address, a register address byte, and a data byte for a single data update. The device acknowledges receipt of each byte by pulling the SDA line low during the high period of a single clock pulse. A valid PMBus address selects the MP4201. The device performs an update on the falling edge of the LSB byte.

PMBus Bus Message Format

In the tables in Figure 14, unshaded cells indicate that the bus host is driving the bus actively, and shaded cells indicate that the MP4201 is driving the bus.

- S = Start condition
- Sr = Repeated start condition
- P = Stop condition
- R = Read bit
- $\overline{W}$ = Write bit c
- A = Acknowledge bit (0)
- $\overline{A}$ = Acknowledge bit (1)

“A” represents the acknowledge bit. The ACK bit is typically active low (logic 0) if the transmitted byte is received successfully by a device.

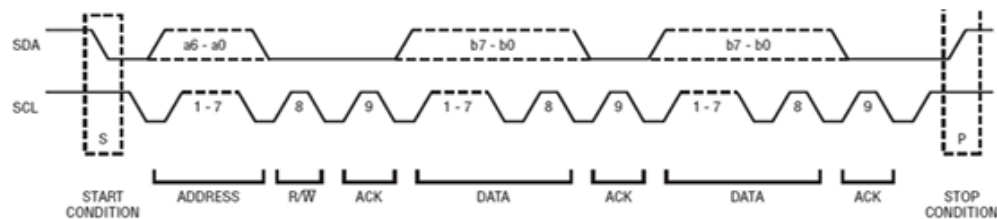
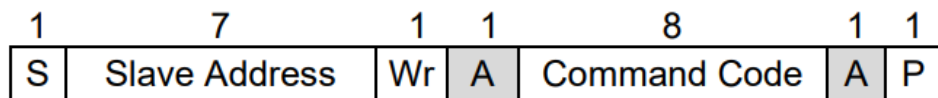


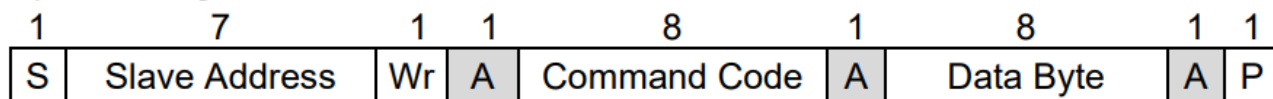
Figure 13: Data Transfer Over the PMBus

MP4201寄存器配置

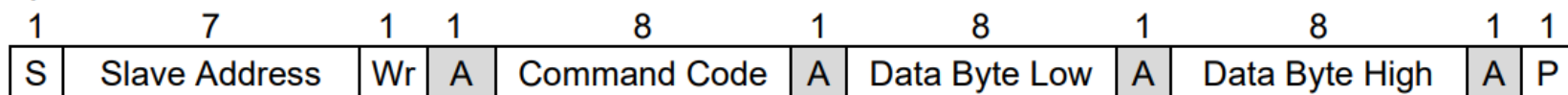
a) Send Byte



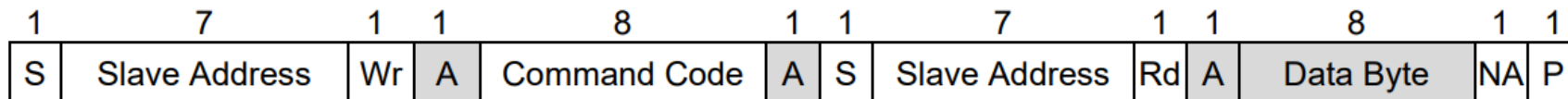
b) Write Byte



c) Write Word



d) Read Byte



e) Read Word

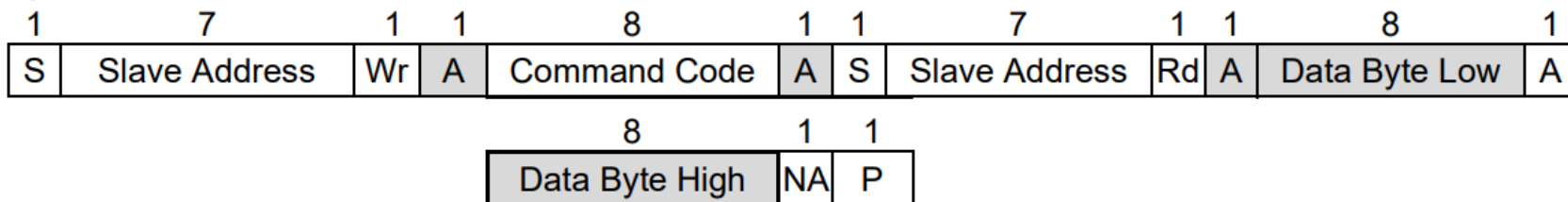


Figure 14: PMBus Message Format

MP4201寄存器配置

REGISTER DESCRIPTION

I2C/PMBUS Register

The I2C is active once VIN/OUT and EN both are higher than UVLO threshold. The register default value is based on MPQ4201-0000.

Command Code	Command Name	Description	Type	Bytes	Data Format	OTP	Default Value
01h	OPEARTION	ON/OFF	R/W	1	Reg	Y	OFF
03h	CLEAR_FAULTS		Send	1	Reg	N	
21h	VOUT_COMMAND	Reference LSB=0.625mV	R/W	2	Direct	Y	12V
46h	IOUT_OC_FAULT_LIMIT	Set output CC current limit continuously, 50mA/step	R/W	2	Direct	Y	10A
59h	VIN_REG_THLD	Reference LSB=0.625mV	R/W	2	Direct	Y	
5Bh	IIN_OC_FAULT_LIMIT	Set input CC current limit continuously, 50mA/step	R/W	2	Direct	Y	10A
79h	STATUS_WORD		R	2	Reg	N	
7Dh	STATUS_TEMPERATURE		R	1	Reg	N	
88h	READ_VIN		R	2	Direct	N	
89h	READ_IIN		R	2	Direct	N	
8Bh	READ_VOUT		R	2	Direct	N	
8Ch	READ_IOUT		R	2	Direct	N	
8Dh	READ_TEMPERATURE		R	2	Direct	N	
D0h	MFR_CTRL1		R/W	1	Reg	Y	
D1h	MFR_CTRL2	Set line drop compensation, GPIO1 and discharge selection	R/W	1	Reg	Y	
D2h	MFR_OCP_CTRL		R/W	1	Reg	Y	
D3h	MFR_CTRL3		R/W	1	Reg	Y	
D4h	MFR_CTRL4	Battery charge control.	R/W	1	Reg	Y	
D5h	MFR_CTRL5	FB mode, slew rate, input UVLOs	R/W	1	Reg	Y	
D6h	MFR_OVP_CTRL	VOUT OVP, and Vin regulation	R/W	1	Reg	Y	
D7h	MFR_PRE_CURRENT	Pre-charge current selection.	R/W	1	Reg	Y	
D8h	MFR_STATUS_MASK	Mask the ALT# pin indication	R/W	1	Reg	Y	
D9h	MFR_OTP_CONFIGURATION_CODE	OTP configuration code	R/W	1	Reg	Y	
DAh	MFR_OTP_REVISION_NUMBER	OTP software revision	R/W	1	Reg	Y	

MP4201寄存器配置

PMBUS COMMAND INTRODUCTION

OPERATION (01h)

Format: Unsigned binary

The OPERATION command is used to configure the operational state of the converter. Bold font is the default.

Bits	Access	Bit Name	Default	Description
7:0	R/W	ON/OFF	8'b 0000 0000	The "EN" pin has higher control priority than this bit. 8'b 0000 0000: The output is off 8'b 1000 0000: The output is on

Default Operation value is set both by I2C/OTP and ADDR pin. Either OTP=80h or ADDR sets EN on, the chip will auto turn-on output. After power up, operation bit can be changed by i2c.

CLEAR_FAULTS (03h)

The CLEAR_FAULTS command is used to clear any fault bits that have been set. This command clears all bits in all status registers simultaneously. At the same time, the device negates (clears, releases) its ALT# signal output if the device is asserting the ALT# signal.

The CLEAR_FAULTS command does not cause a unit that has latched off for a fault condition to restart.

If the fault is still present when the bit is cleared, the fault bit shall immediately be set again and the host notified by the usual means. **This command is write only.** Refer to Send Byte format.

VOUT_COMMAND (21h)

Format: Unsigned binary

The VOUT_COMMAND register value is to set the output voltage in both charging mode and discharging mode.

When power flow is from VIN to OUT, the VOUT_COMMAND sets the internal FB reference and regulation voltage of OUT pin.

When power flow is from OUT to VIN, the VOUT_COMMAND sets the internal FB reference and regulation voltage of VIN pin.

Command	VOUT_COMMAND															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Function	-				V											
Default value	600 Decimal (12V) FB Reference LSB=0.625mV															

The Vout, in volts is calculated from the equation below:

$$V_{out} (V) = V \times 0.625 \times \text{Feedback Ratio} / 1000 \quad (2)$$

Where:

V is a 12 bit unsigned binary integer of VOUT_COMMAND [11:0]. 0.625mV/step.

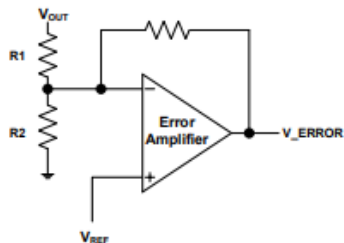
The valid internal reference voltage range is from 0.1V to 2.56V with 0.625mV step (total 4096 steps, 12 bit DAC). Don't suggest the FB reference voltage is below 0.1V.

When use internal feedback resistor, the feedback ratio is fixed at 32, the valid Vout range is from 3.2V to 81.92V. If the output voltage requirement is >81.92V, external feedback divider resistors should be used.

MP4201寄存器配置

Internal FB Ratio	The Regulation Voltage Range
Internal Feedback Ratio=32	3.2V to 81.92V

When use the external divider resistor, the feedback ratio is defined as $(R1+R2) / (R2)$.



VIN_REG_THLD (59h)

Format: Unsigned binary

The VIN_REG sets the input voltage regulation threshold voltage to limit the output power for both direction.

When power flow is from VIN to OUT, the VIN_REG_THLD sets the VIN pin regulation voltage.

When power flow is from OUT to VIN, the VIN_REG_THLD sets the OUT pin regulation voltage.

Command	VIN_REG_THLD															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Function	-				V											
Default value	1200 Decimal FB Reference LSB=0.625mV															

The Vin regulation voltage, in volts is calculated from the equation below:

$$Vin_Reg (V) = (V \times 0.625) \times \text{Feedback Ratio} / 1000 \quad (3)$$

Where:

V is a 12 bit unsigned binary integer of VIN_REG [11:0].

The internal feedback ratio is 32, the maximum VIN regulation voltage is 81.92V, 20mV/step. The Minimum input regulation voltage should be higher than the UVLO.

IOUT_OC_FAULT_LIMIT (46h)

Format: Unsigned binary

Set the output constant current (CC) limit threshold.

When power flow is from OUT to VIN, the IOUT_OC_FAULT_LIMIT sets the IS1P and IS1N current limit.

When power flow is from VIN to OUT, the IOUT_OC_FAULT_LIMIT sets the IS2P and IS2N current limit.

MP4201寄存器配置

Command	IOUT_OC_FAULT_LIMIT															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Function	-							A								
Default value	200 Integer (10A)															

The real value I_{OUT_OC} , in Ampere is calculated from the equation below:

$$I_{OUT_OC} (A) = I_{OUT_LIM} * 0.05 \quad (4)$$

Where:

I_{OUT_LIM} is a 9 bit unsigned binary integer of $I_{OUT_LIM} D[8:0]$;

The I_{OUT_OC} resolution or minimum step is 50mA. Maximum value is 25A. The current sense resistor should be the same as I2C setup value in register. If the current sense resistor doesn't match the I2C setup value, the real output current limit may be different with setup value. For example, if the real current sense resistor is 2mΩ in circuitry, but the I2C setup value is 5mΩ, the output current limit should be 2.5× I_{OUT_OC} value.

IIN_OC_FAULT_LIMIT (5Bh)

Format: Unsigned binary

Set the input constant current (CC) limit threshold.

When power flow is from VIN to OUT, the IIN_OC_FAULT_LIMIT sets the IS1P and IS1N current limit.

When power flow is from OUT to VIN, the IIN_OC_FAULT_LIMIT sets the IS2P and IS2N current limit.

Command	IIN_OC_FAULT_LIMIT															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Function	-							A								
Default value	200 Integer (10A)															

The real value I_{IN_OC} , in Ampere is calculated from the equation below:

$$I_{IN_OC} (A) = I_{IN_LIM} * 0.05 \quad (5)$$

Where:

I_{IN_LIM} is a 9 bit unsigned binary integer of $I_{IN_LIM} D[8:0]$;

The I_{IN_OC} resolution or minimum step is 50mA. Maximum value is 25A. As the input current sense resistor is a part of the loop control, and it also sets the cycle-by-cycle FETs peak current limit, **the input current sense resistor must be same as I2C setup value in register.**

STATUS_WORD (79h)

Format: Unsigned binary

The STATUS_WORD command returns two bytes of information with a summary of the unit's fault condition. Based on the information in these bytes, the host can get more information by reading the appropriate status registers

MP4201寄存器配置

Byte	Bit	Bit Name	Behavior	Description
High	15	RESERVED	N/A	Reserved. Default value is 0b.
	14	SCP_FAULT	Latch	An output short circuits fault has occurred. When output current limit is triggered and VOUT UV is triggered, this bit will be set 1.
	13	INPUT_OV_FAULT	Latch	An input over voltage fault occurs.
	12	RESERVED	N/A	Reserved. Default value is 0b.
	11	PG_FAULT#	Live	The POWER_GOOD signal, if present, is negated. When this bit is set '1', the output voltage is not good. When this bit is clear, the output voltage is power good.
	10	RESERVED	N/A	Reserved. Default value is 0b.
	9	Charge Complete	Live	When this bit is set 1, the device charging is complete. The charging current is below than the termination current limit setup.
	8	IIN_OC_FAULT	Latch	An input over current fault has occurred. Once enter CC current limit, this bit will be set.
Low	7	RESERVED	N/A	Reserved. Default value is 0b.
	6	RESERVED	N/A	Reserved. Default value is 0b.
	5	VOUT_OV_FAULT	Latch	An output over voltage fault has occurred.
	4	IOUT_OC_FAULT	Latch	An output over current fault has occurred. Either enter CC current limit, Peak current limit or enter hiccup mode, this bit will be set.
	3	RESERVED	N/A	Reserved. Default value is 0b.
	2	TEMPERATURE	Latch	A temperature warning or fault has occurred including die temperature sense and NTC function.
	1	RESERVED	N/A	Reserved. Default value is 0b.
	0	CRC_Error	Latch	A CRC error has occurred including user otp or analog trim.

The exception to the rule that status bit remain set is the PG_FAULT# bits. This bit always reflects the current state of the POWER_GOOD signal (if present).

STATUS_TEMPERATURE (7Dh)

Format: Unsigned binary

The STATUS_TEMPERATURE command returns one data byte with contents as follows:

Bit	Description
7	OT_FAULT (Over temperature Fault). OTP entry threshold 165°C with 20°C hysteresis.
6	OT_WARNING (Over temperature warning). OTP entry threshold 135°C with 20°C hysteresis.
5	NTC_FAULT is triggered.
4	RESERVED
3	
2	
1	
0	

MP4201寄存器配置

READ_VIN (88h)

Format: Unsigned binary

The READ_VIN command returns the measured input voltage (V_{IN}).

When power flow is from VIN to OUT, the READ_VIN represents the VIN pin voltage.

When power flow is from OUT to VIN, the READ_VIN represents the OUT pin voltage.

Command	READ_VIN															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Function	-							Y								

READ_VIN (in V) can be calculated with Equation (8):

$$V_{IN} (V) = Y \times 0.0832 \quad (6)$$

Where Y is an 10-bit unsigned binary integer of READ_VIN[9:0], and the internal V_{IN} ADC sample resolution is 83.2mV/LSB.

For the most accurate reading, it is recommended to take the average of 5 or more readings.

READ_IIN (89h)

Format: Unsigned binary

The READ_IIN command returns the measured output current (I_{IN}).

When power flow is VIN to OUT, the READ_IIN represents the IS1P and IS1N current.

When power flow is from OUT to VIN, the READ_IIN represents the IS2P and IS2N current.

Command	READ_IIN															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Function	-							Y								

READ_IIN (in A) can be calculated with Equation (7):

$$I_{IN} (A) = Y \times 0.043 \quad (7)$$

Where Y is an 10-bit unsigned binary integer of READ_IIN[9:0], and the internal current sense ADC sample resolution is 43mA / LSB. The real current sense resistor should match the register value setup, otherwise, the read out value is not correct.

For the most accurate reading, it is recommended to take the average of 5 or more readings.

READ_VOUT (8Bh)

Format: Unsigned binary

The READ_VOUT command returns the measured output voltage (V_{OUT}).

When power flow is from OUT to VIN, the READ_VOUT represents the VIN pin voltage.

When power flow is from VIN to OUT, the READ_VOUT represents the OUT pin voltage.

Command	READ_VOUT															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Function	-							Y								

MP4201寄存器配置

READ_VOUT (in V) can be calculated with Equation (8):

$$V_{OUT} (V) = V \times 0.0832 \quad (8)$$

Where V is a 10-bit unsigned binary integer of READ_VOUT[9:0], and the internal feedback ADC sample resolution is 83.2mV/LSB.

In automatic pulse-frequency modulation (PFM), the user should send the READ_VOUT command twice to receive the updated V_{OUT} value. For the most accurate reading, it is recommended to take the average of 5 or more readings.

READ_IOUT (8Ch)

Format: Unsigned binary

The READ_IOUT command returns the measured output current (I_{OUT}).

When power flow is from OUT to VIN, the READ_IOUT represents the IS1P and IS1N current.

When power flow is from VIN to OUT, the READ_IOUT represents the IS2P and IS2N current.

Command	READ_IOUT															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Function	-							Y								

READ_IOUT (in A) can be calculated with Equation (7):

$$I_{OUT} (A) = Y \times 0.043 \quad (9)$$

Where Y is an 10-bit unsigned binary integer of READ_IOUT[9:0], and the internal current sense ADC sample resolution is 43mA / LSB. The real current sense resistor should match the register value setup, otherwise, the read out value is not correct.

For the most accurate reading, it is recommended to take the average of 5 or more readings.

READ_TEMPERATURE (8Dh)

Format: Unsigned binary

The READ_TEMPERATURE command returns the IC's silicon temperature.

Command	READ_TEMPERATURE															
Format	Direct															
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Access	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
Function	-							Y								

READ_TEMPERATURE (in °C) can be calculated with Equation (10):

$$\text{Temperature } (^\circ\text{C}) = -272.48 + 1.07 \times Y \quad (10)$$

Where Y is an 10-bit two's complement integer of READ_TEMPERATURE[9:0]

For the most accurate reading, it is recommended to take the average of 5 or more readings.

MP4201寄存器配置

I2C Register Map

Name	REG (0x)	R/W	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
MFR_CTRL1	D0	R/W	DIR	INPUT UVLO SEL		-	FB Mode	OC MODE	Dead Time	
MFR_CTRL2	D1	R/W	-		Discharge EN	Discharge Current		Line Drop Compensation		
MFR_OCP_CTRL	D2	R/W	Switching Current Limit			-	Rsens 2		Rsens 1	
MFR_CTRL3	D3	R/W	OUT_C C_EN	IN_CC _EN	ADC Force On	GPIO		I2C Slave Address(A3-A1)		
MFR_CTRL4	D4	R/W	Battery Charge EN	Input CC Blank Timer		Pre-Charge EN	Output CC Blank Timer		Pre-Charge Voltage	EN_TER M_CHG
MFR_CTRL5	D5	R/W	Slew Rate Rise		Slew Rate Fall		Termination Current, 200mA/step			
MFR_OVP_CTRL	D6	R/W	VIN_RE G_EN	ABS VOUT OVP			VOUT OVP EN	-	VIN OVP	
MFR_PRE_CURRENT	D7	R/W	-	Pre-charge current, 50mA/step						
MFR_STATUS_MASK	D8	R/W	Mask the ALT# pin indication when fault or event happens							
MFR_OTP_CONFIGURATION CODE	D9	R/W	OTP configuration code, defined by MPS							
MFR_OTP_REVISION_NUMBER	DA	R/W	OTP software revision number, defined by MPS							

I2C Slave "ADDRESS"

The I2C slave address:

The address A3-A1 bits are set by MFR_CTRL3_D[2:0], the A5-A4 bit is set by ADDR pin status. A7-A6 bits are fixed 01b.

I2C Address A7:A1, 01xxxx			Default OPERATION
GPIO_ADDR Status	Binary	Hex	
ADDR=GND	010 0111	27H	OFF
ADDR=0.3×VCC	010 1111	2FH	OFF
ADDR=0.6×VCC	011 0111	37H	ON
ADDR=VCC	011 1111	3FH	ON

MFR_CTRL1 (D0h)

Format: Unsigned binary

The function controlled by MFR_CTRL1 register is shown in below table.

Bits	Access	Bit Name	Default	Description
7	R/W	DIR	1'b 0	Set buck-boost working direction. Both of the DIR bit and DIR pin status can determine the working direction. If the power direction need be changed, the EN pin should be pulled low or set I2C EN=0 (0x01=0), then DIR logic can be changed. 1'b 0: The current flow is from OUT pin to VIN pin @DIR pin=low 1'b 1: The current flow is from VIN pin to OUT pin no matter DIR pin status
6:5	R/W	INPUT_UVLO_SEL	2'b 00	Input UVLO threshold selection. The switching is disabled when the VIN in charging mode or OUT voltage in discharge mode is below the input UVLO threshold. 2'b 00: Minimum UVLO, refer to EC table 2'b 01: 8.5V Rising Voltage 2'b 10: 11V Rising Voltage 2'b 11: 20V Rising Voltage

MP4201寄存器配置

4	R/W	RESERVED	N/A	Reserved.
3	R/W	FB_MODE	1'b 1	Select the output feedback path. 1'b 0: External Feedback Divider Mode through FB pin 1'b 1: Internal Feedback, the feedback ratio is fixed at 32
2	R/W	OCP_MODE	1'b 0	Set input and output OCP protection mode after triggering the current limit for both directions. 1'b 0: Hiccup Mode 1'b 1: Constant Current Mode. The inductor current is limited by constant current limit or switching current limit.
1:0	R/W	DEAD_TIME	2'b 01	SWx dead time selection. 2'b 00: 15ns 2'b 01: 20ns 2'b 10: 25ns 2'b 11: 30ns

MFR_CTRL2 (D1h)

Format: Unsigned binary

The function controlled by MFR_CTRL2 register is shown in below table.

Bits	Access	Bit Name	Default	Description
7:6	R	RESERVED	N/A	Reserved.
5	R/W	DISCHARGE_EN	1'b 1	The output discharge enable or disable control bit. 1'b 0: the output discharge is disabled 1'b 1: the output discharge is enabled
4:3	R/W	DISCHARGE_CURRENT	2'b 01	Output discharge current selection. 2'b 00: 25mA 2'b 01: 50mA (default), suggest output voltage is <60V 2'b 10: 75mA, suggest output voltage range is <40V 2'b 11: 100mA, suggest output voltage range is <25V
2:0	R/W	LINE_DROP_COMPENSATION	3'b 000	Set output voltage compensation value vs. load current. When set in discharging mode, the line drop compensation is active on VIN pin; when set in charging mode, the line drop compensation is active on OUT pin. 3'b 000: no compensation 3'b 001: Vout compensates 80mV@5A Iout 3'b 010: Vout compensates 160mV@5A Iout 3'b 011: Vout compensates 240mV@5A Iout 3'b 100: Vout compensates 360mV@5A Iout 3'b 101: Vout compensates 480mV@5A Iout 3'b 110: Vout compensates 600mV@5A Iout 3'b 111: Vout compensates 720mV@5A Iout Above compensation amplitude is fixed for any output voltage when feedback ratio is 32. If external feedback is used, actual output voltage compensation is register's value multiple FB ratio/32.

MP4201寄存器配置

MFR_OCP_CTRL (D2h)

Format: Unsigned binary

The function controlled by MFR_OCP_CTRL register is shown in below table.

Bits	Access	Bit Name	Default	Description
7:5	R/W	SWITCHING_CUR RENT_LIMIT	3'b 011	Set the peak and valley current limit 3'b 000: 10A peak current limit/ 8A valley current limit 3'b 001: 15A peak current limit/ 12A valley current limit 3'b 010: 20A peak current limit/ 17A valley current limit 3'b 011: 25A peak current limit/ 22A valley current limit 3'b 100: 30A peak current limit/ 26A valley current limit 3'b 101: 35A peak current limit/ 30A valley current limit
4	R	RESERVED	N/A	Reserved.
3:2	R/W	RSSENS_2	2'b 00	Select the current sense resistor value across IS2P and IS2N. 2'b 00: 5mΩ 2'b 01: 2mΩ
1:0	R/W	RSSENS_1	2'b 00	Select the current sense resistor value across IS1P and IS1N. 2'b 00: 5mΩ 2'b 01: 2mΩ

MFR_CTRL3 (D3h)

Format: Unsigned binary

The function controlled by MFR_CTRL3 register is shown in below table.

Bits	Access	Bit Name	Default	Description
7	R/W	OUT_CC_EN	1'b 1	Output average current control bit. 1'b 0: Output CC current limit is disabled 1'b 1: Output CC current limit is enabled
6	R/W	IN_CC_EN	1'b 0	Input average current control bit. 1'b 0: Input CC current limit is disabled 1'b 1: Input CC current limit is enabled.
5	R/W	ADC_FORCED_O N	2'b 0	The ADC function enable/disable control when set I2C Operation off. 2'b 0: ADC function is disabled 2'b 1: ADC function is enabled
4:3	R/W	GPIO	2'b 00	GPIO function select pin: 2'b 00: PG Indication 2'b 01: External clock synchronization 2'b 10: Battery charging complete indication.
2:0	R/W	I ² C ADDRESS	3'b 111	Set I2C slave address A3-A1 bit. Default I2C slave address is 27H when ADDR pin=GND

MP4201寄存器配置

MFR_CTRL4 (D4h)

Format: Unsigned binary

The function controlled by MFR_CTRL4 register is shown in below table.

Bits	Access	Bit Name	Default	Description
7	R/W	BATTERY_CHARGE_EN	1'b 0	Battery charge function enable control bit including pre-charge and termination detection. 1'b 0: the battery charge function is disabled including pre-charge and termination detection 1'b 1: the battery charge function is enabled including pre-charge, CC charge and termination detection
6:5	R/W	INPUT_CC_BLANK_TIMER	2'b 00	Set the blank time before enters CC mode: 2'b 00: 400μs 2'b 01: 4ms 2'b 10: 16ms 2'b 11: 64ms Default is "00". Set the CC blank timer to 100μs.
4	R/W	PRE-CHARGE_EN	1'b 1	The pre-charge cycle control bit. 1'b 0: the pre-charge is disabled 1'b 1: the pre-charge is enabled
3:2	R/W	OUTPUT_CC_BLANK_TIMER	2'b 00	Set the blank time before enters CC mode: 2'b 00: 400μs 2'b 01: 4ms 2'b 10: 16ms 2'b 11: 64ms Default is "00". Set the CC blank timer to 100μs.
1	R/W	PRE-CHARGE_VOLTAGE	3'b 0	Battery charge threshold setup from pre-charge to fast charge, the hysteresis voltage is 1.5% of VOUT_COMMAND voltage. 1'b 0: 60% of VOUT_COMMAND 1'b 1: 70% of VOUT_COMMAND
0	R/W	EN_TERM_CHG	1'b 0	The termination charge control bit. 1'b 0: The charge will not be disabled when $I_{CHG} < I_{TERM}$ 1'b 1: The charge will be disabled and charge is terminated when $I_{CHG} < I_{TERM}$

MFR_CTRL5 (D5h)

Format: Unsigned binary

The function controlled by MFR_CTRL5 register is shown in below table.

Bits	Access	Bit Name	Default	Description
7:6	R/W	SLEW_RATE_RISE	2'b 01	Set the Vout adjust rising slew rate. 2'b 00: 0.08mv/us Vref rising slew rate 2'b 01: 0.16mv/us Vref rising slew rate 2'b 10: 0.4mv/us Vref rising slew rate 2'b 11: 0.8mv/us Vref rising slew rate Vout_Slew rate=Vref_slew rate * Feedback Ratio.

MP4201寄存器配置

5:4	R/W	SLEW_RATE_FALL	2'b 10	Set the Vout adjust falling slew rate. 2'b 00: 0.005mv/us Vref falling slew rate 2'b 01: 0.04mv/us Vref falling slew rate 2'b 10: 0.1mv/us Vref falling slew rate 2'b 11: 0.2mv/us Vref falling slew rate Vout_Slew rate=Vref_slew rate * Feedback Ratio.
3:0	R/W	TERMINATION_CURRENT	3'b 001	The battery charging termination charge current selection. The current sense resistor should match the register setup, otherwise the charging current value will be changed linearly. Default is 0x01, 400mA. 200mA/step.

MFR_OVP_CTRL (D6h)

Format: Unsigned binary

The function controlled by MFR_OVP_CTRL register is shown in below table.

Bits	Access	Bit Name	Default	Description
7	R/W	VIN_REG_EN	1'b 0	VIN regulation enable control. 1'b 0: The VIN regulation function is disabled 1'b 1: The VIN regulation function is enabled. The output power will be limited to avoid the input voltage below than VIN_REG_THLD setup value
6:4	R/W	ABS_VOUT_OVP	3'b 101	Output absolute over voltage threshold selection, the hysteresis voltage is 1.6V. 3'b 000: 30.8V 3'b 001: 39.6V 3'b 010: 52.8V 3'b 011: 62V 3'b 100: 84V 3'b 101: 104V
3	R/W	VOUT_OVP_EN	1'b 1	Output OVP enable or disable control bit. 1'b 0: no output OVP function, both FB OVP and absolute OVP are disabled. 1'b 1: enable, both FB OVP and absolute OVP are enabled
2	R	RESERVED	N/A	Reserved.
1:0	R/W	VIN_OVP	2'b 11	The input OVP threshold selection. Input OVP has 1.5V hysteresis. 2'b 00: 84V 2'b 01: 30V 2'b 10: 60V 2'b 11: 104V

MP4201寄存器配置

MFR_PRE_CURRENT (D7h)

Format: Unsigned binary

The battery pre-charge current selection. The current sense resistor should match the register setup, otherwise the charging current value will be changed linearly.

Command	Pre-charge current							
format	Direct							
Bit	7	6	5	4	3	2	1	0
Access	-	r/w	r/w	r/w	r/w	r/w	r/w	r/w
Function	-	0.1A-6.4A, 50mA/step						
Default value	-	0x10, 800mA.						

MFR_STATUS_MASK (D8h)

Format: Unsigned binary

This register only can mask off the ALT# pin behavior, but the STATUS_WORD register still indicate each event.

Bits	Access	Bit Name	Default	Description
7	R/W	VOUT_MSK	1'b 1	This bit mask output OV fault. 0: Not mask 1: Mask Enabled
6	R/W	IOUT/POUT_MSK	1'b 0	This bit mask IOUT_OC_FAULT, IOUT/POUT and OC_EXIT. 0: Not mask. 1: Mask Enabled
5	R/W	SCP_FAULT_MSK	1'b 1	This bit mask the SCP fault. 0: Not Mask 1: Mask Enabled
4	R/W	TEMP_MSK	1'b 1	Temperature related mask bit including OTW, OTP and NTC function. 0: Not mask 1: Mask Enabled
3	R/W	PG_STATUS#_MSK	1'b 1	Higher level PG mask off control bit. 0: Not mask 1: Mask Enabled
2	R/W	CHARGE_COMPLETE_MSK	1'b 1	Battery charging complete mask off control. 0: Not mask 1: Mask Enabled
1	R/W	VIN_MSK	1'b 1	This bit mask INPUT OV FAULT. 0: Not mask 1: Mask Enabled
0	R/W	IIN_MSK	1'b 1	This bit mask the IIN_OC_FAULT. 0: Not mask. 1: Mask Enabled

MP4201寄存器配置

MFR_OTP_CONFIGURATION_CODE (D9h)

Format: Unsigned binary

Bits	Bit Name	Description
7:0	OTP_CONFIGURATION_CODE	OTP configuration code, defined by MPS.

Registers "MFR_OTP_REVISION_NUMBER (DAh)

Format: Unsigned binary

Bits	Bit Name	Description
7:0	OTP_REVISION_NUMBER	OTP software revision number, defined by MPS

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MP4201应用指南

APPLICATION INFORMATION

Output Voltage Setting

The default output voltage is set using a resistor divider to FB. The bottom resistor in the resistor divider is typically in the 1kΩ to 50kΩ range.

The top resistor in the feedback resistor divider is selected using Equation (11):

$$R1 = \frac{V_{OUT} - V_{REF}}{V_{REF}} \times R2 \quad (11)$$

It is possible to use the I²C interface to select the FB V_{REF} and get another output voltage.

Inductor Selection

The inductor selection is based on the work mode. The inductance for the buck mode is calculated with Equation (12):

$$L_{BUCK} = \frac{V_{OUT}}{f_{SW} \times \Delta I_L} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \quad (12)$$

Where ΔI_L is the peak-to-peak inductor ripple current, to balance the inductor size and thermal rise.

In boost mode, the target inductance for boost mode is calculated with Equation (13) and Equation (14):

$$L_{BOOST} = \frac{V_{IN} \times (V_{OUT} - V_{IN})}{V_{OUT} \times f_{SW} \times \Delta I_L} \quad (13)$$

$$I_{IN(MAX)} = \frac{V_{OUT} \times I_{LOAD(MAX)}}{V_{IN} \times \eta} \quad (14)$$

Where $I_{LOAD(max)}$ is the maximum load current, ΔI_L is the peak-to-peak ripple current, and η is the efficiency.

Choosing a larger inductance reduces the ripple current but also increases the size of the inductor and reduces the achievable bandwidth of the converter by moving the right half-plane zero to lower frequencies. The appropriate balance should be chosen based on the application requirements. ΔI_L can be 2A ~ 8A according to the inductor size vs. thermal performance.

Max inductance in boost mode is limited by the right half-plane zero (RHPZ). Generally, $F_{RHPZ} > 3 \times F_{BW}$ to keep loop stability. F_{BW} is

considered as 10kHz in most cases. L_{max} can be estimated with Equation (15)

$$L_{MAX} = \frac{R_{LOAD}}{2 \times \pi \times F_{RHPZ}} \times \left(\frac{V_{IN_MIN}}{V_{OUT}}\right)^2 \quad (15)$$

Input Capacitor Selection

In buck mode, the MP4201 has a discontinuous input current (boost mode is continuous), and requires a capacitor to supply the AC current during buck mode while maintaining the DC input voltage. Ceramic capacitors are recommended for best performance, and should be placed as close to VIN as possible. Capacitors with X5R or X7R ceramic dielectrics are recommended because of their stable temperature characteristics. The capacitors must also have a ripple current rating greater than the maximum input ripple current of the converter. The buck mode input ripple current can be estimated with Equation (16):

$$I_{CIN_RMS} = I_{OUT} \times \sqrt{\frac{V_{OUT}}{V_{IN}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right)} \quad (16)$$

The worst-case condition in buck mode occurs at $V_{IN} = 2V_{OUT}$, calculated with Equation (17):

$$I_{CIN_RMS} = \frac{I_{OUT}}{2} \quad (17)$$

For simplification, choose an input capacitor with an RMS current rating greater than half of the maximum load current.

The input capacitance value determines the input voltage ripple of the converter. If there is an input voltage ripple requirement in the system, choose an input capacitor that meets the specification.

In buck mode, the input voltage ripple can be estimated with Equation (18):

$$\Delta V_{IN} = \frac{I_{OUT}}{f_{SW} \times C_{IN}} \times \frac{V_{OUT}}{V_{IN}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \quad (18)$$

The worst-case condition occurs at $V_{IN} = 2V_{OUT}$, calculated with Equation (19):

$$\Delta V_{IN} = \frac{1}{4} \frac{I_{OUT}}{f_{SW} \times C_{IN}} \quad (19)$$

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The input voltage ripple in boost mode can be estimated:

$$\Delta V_{IN} = \frac{V_{IN}}{8 \times f_{SW}^2 \times L \times C_{IN}} \times \left(1 - \frac{V_{IN}}{V_{OUT}}\right) \quad (20)$$

If using polymer, hybrid, or low-ESR electrolytic capacitors, the ESR dominates the impedance at the switching frequency. The input voltage ripple in boost mode can be estimated with Equation:

$$\Delta V_{IN} = \frac{V_{IN}}{f_{SW} \times L} \times \left(1 - \frac{V_{IN}}{V_{OUT}}\right) \times \left(R_{ESR} + \frac{1}{8 \times f_{SW} \times C_{IN}}\right) \quad (21)$$

Output Capacitor Selection

In boost mode, the output current is discontinuous, so C_{OUT} must be capable of reducing the output voltage ripple.

A higher capacitance value may be required to lower the output ripple and the transient response. Low-ESR capacitors, such as X5R or X7R ceramic capacitors, are recommended. If using ceramic capacitors, the capacitance dominates the impedance at the switching frequency, so the output voltage ripple is independent of the ESR. The output voltage ripple in boost mode is estimated with Equation (22):

$$\Delta V_{OUT} = \frac{\left(1 - \frac{V_{IN}}{V_{OUT}}\right) \times I_{LOAD}}{C_{OUT} \times f_{SW}} \quad (22)$$

The output voltage ripple in buck mode is estimated with Equation (23):

$$\Delta V_{OUT} = \frac{V_{OUT}}{8 \times f_{SW}^2 \times L \times C_{OUT}} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \quad (23)$$

If using hybrid, polymer, or low-ESR electrolytic capacitors, the ESR dominates the impedance at the switching frequency, so the output ripple in boost mode is estimated using Equation (24):

$$\Delta V_{OUT} = \frac{\left(1 - \frac{V_{IN}}{V_{OUT}}\right) \times I_{LOAD}}{C_{OUT} \times f_{SW}} + \frac{I_{LOAD} \times R_{ESR} \times V_{OUT}}{V_{IN}} \quad (24)$$

The output voltage ripple in buck mode is estimated with Equation (25):

$$\Delta V_{OUT} = \frac{V_{OUT}}{f_{SW} \times L} \times \left(1 - \frac{V_{OUT}}{V_{IN}}\right) \times \left(R_{ESR} + \frac{1}{8 \times f_{SW} \times C_{OUT}}\right) \quad (25)$$

Where R_{ESR} is the equivalent series resistance of the output capacitors.

Choose output capacitors to satisfy the output ripple and load transient requirements of the design. Capacitance derating should be taken into consideration when designing high output voltage applications.

External MOSFET Selection

The MP4201 requires four external N-channel power MOSFETs. Figure 13 shows two for the top switches (switches A and D) and two for the bottom switches (switches B and C). In buck mode, SWA and SWB switch while SWD remains on. In boost mode, SWC and SWD switch while SWA remains on.

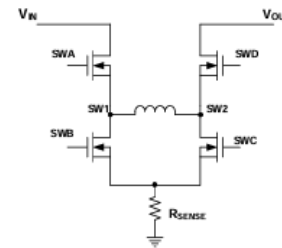


Figure 13: Buck-Boost Topology

The critical parameters of selecting a MOSFET are:

1. Maximum drain-to-source voltage, $V_{DS(MAX)}$
The SWA and SWB need to withstand the maximum input voltage and the transient spikes at SW1 during switching. Therefore, it is recommended to select $V_{DS(MAX)}$ for SWA and SWB at 1.5 times the input voltage.
The SWC and SWD see output voltage and transient spikes at SW2 during switching. Therefore, it is recommended to select SWC and SWD at ≥ 1.5 times the output voltage.
2. Maximum continuous drain current, $I_{D(MAX)}$
The SWA, SWB, SWC and SWD need to withstand the maximum inductor current and the peak inductor current during switching.
3. V_{TH} : The driver voltages of the MP4201 are supplied by VCC. **The recommended maximum gate-to-source V_{th} is 3.5V.** The gate plateau voltages of the MOSFETs should be smaller than minimum Vcc voltage of the converter, otherwise the MOSFETs

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may not fully enhance during start-up or overload conditions.

4. Drain-source on-state resistance, $R_{DS(ON)}$

Lower $R_{DS(ON)}$ is better for MOSFETs thermal rise management as well as obtaining higher efficiency. Refer to the calculation of conduction and switching loss in section below for $R_{DS(ON)}$ selection.

MOSFET SWA

When the MP4201 works in boost mode, SWA is on consistently. Its conduction power loss can be calculated with Equation (26):

$$P_{C_Loss(SWA)} = (I_o \times \frac{V_{OUT}}{V_{IN}})^2 \times R_{DS(ON)(SWA)} \quad (26)$$

Assume that the MOSFET junction-to-ambient thermal resistance is 40°C/W (this is determined by the board power dissipation), and that the maximum acceptant temperature rise is 80°C , thus, the maximum power loss is 2W, shown with Equation (27):

$$P_{Loss} < 2W \quad (27)$$

Based on this equation, we can select the MOSFET R_{ON} .

When the MP4201 works in buck mode, the conduction and switching loss of SWA can be calculated with Equation (28) and Equation (29), respectively:

$$P_{C_Loss(SWA)} = \frac{V_{OUT}}{V_{IN}} \times I_o^2 \times R_{DS(ON)(SWA)} \quad (28)$$

$$P_{SW_Loss(SWA)} = \frac{1}{2} V_{IN} \times I_{OUT} \times (t_{on} + t_{off}) \times f_{sw} \quad (29)$$

The switch on time (t_{on}) and the switch off time (t_{off}) are based on the MOSFET datasheet information (see Figure 14).

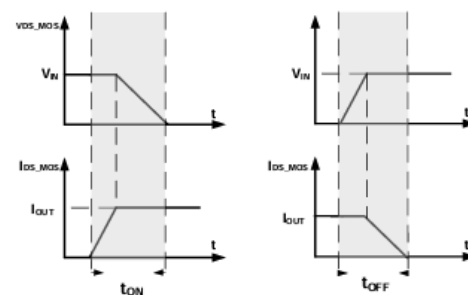


Figure 14: Switch On Time and Switch Off Time

MOSFET SWB

When MP4201 work in buck mode, its conduction loss can be calculated with Equation (30):

$$P_{C_Loss(SWB)} = (1 - \frac{V_{OUT}}{V_{IN}}) \times I_o^2 \times R_{DS(ON)(SWB)} \quad (30)$$

MOSFET SWC

When the MP4201 works in boost mode, SWB is always off. Its conduction loss in boost mode can be calculated with Equation (31):

$$P_{C_Loss(SWC)} = (1 - \frac{V_{IN}}{V_{OUT}}) \times I_o \times \frac{V_{OUT}}{V_{IN}} \times R_{DS(ON)(SWC)} \quad (31)$$

When the MP4201 works in boost mode, the SWC switching loss can be calculated with Equation (32):

$$P_{SW_Loss(SWC)} = \frac{1}{2} \times V_{OUT} \times (I_{OUT} \times \frac{V_{OUT}}{V_{IN}}) \times (t_{on} + t_{off}) \times f_{sw} \quad (32)$$

MOSFET SWD

When the MP4201 works in buck mode, SWD is on consistently. Its power loss can be calculated with Equation (33):

$$P_{C_Loss(SWD)} = I_o^2 \times R_{DS(ON)(SWD)} \quad (33)$$

When the MP4201 works in boost mode, the SWD conduction loss can be calculated with Equation (34):

$$P_{C_Loss(SWD)} = (\frac{V_{IN}}{V_{OUT}}) \times (I_o \times \frac{V_{OUT}}{V_{IN}})^2 \times R_{DS(ON)(SWD)} \quad (34)$$

Dead time and the low-side MOSFET switching loss can be ignored.

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Compensation Components

The COMP pin controls system stability and transient response. COMP is the output of the internal error amplifier. A series capacitor-resistor combination sets a pole-zero combination to control the control system's characteristics.

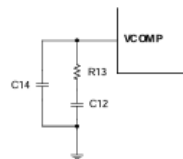


Figure 15: COMP External Compensation

The COMP external compensation sets one pole F_{P1} and one zero F_{Z1} (see Figure 15). These are determined by Equation (35) and Equation (36):

$$F_{P1} = \frac{1}{2\pi \times C14 \times R13} \quad (35)$$

$$F_{Z1} = \frac{1}{2\pi \times C12 \times R13} \quad (36)$$

When the MP4201 works in buck mode, the DC gain of the voltage feedback loop is calculated with Equation (37):

$$A_{VDC} = R_{LOAD} \times \frac{G_{CS}}{R_{SENSE}} \times A_{V-EA} \times \frac{V_{FB}}{V_{OUT}} \quad (37)$$

Where A_{V-EA} is the error-amplifier voltage gain ($712\mu A/V \times R13$), G_{CS} is the COMP to current sense gain ($16.7A/V$), R_{SENSE} is the current sense resistor, and R_{LOAD} is the load resistor value.

The system has two important poles: one is from the compensation capacitor (C10) and the output resistor of the error amplifier, and the other one is from the output capacitor and the load resistor. These poles can be calculated with Equation (38) and Equation (39), respectively:

$$F_{P2} = \frac{G_{EA}}{2\pi \times C12 \times A_{V-EA}} \quad (38)$$

$$F_{P3} = \frac{1}{2\pi \times C_{OUT} \times R_{LOAD}} \quad (39)$$

Where G_{EA} is the error-amplifier transconductance ($712\mu A/V$), and C_{OUT} is the output capacitor.

The system may have another significant zero if the output capacitor has a large capacitance or a high ESR value. This zero can be located with Equation (40):

$$F_{ESR} = \frac{1}{2\pi \times C_{OUT} \times R_{ESR}} \quad (40)$$

When the MP4201 works in boost mode, the DC gain of the voltage feedback loop is calculated with Equation (41):

$$A_{VDC} = \frac{V_{IN} \times A_{V-EA} \times R_{LOAD} \times V_{FB} \times G_{CS} \times R13}{2 \times V_{OUT}^2 \times R_{SENSE}} \quad (41)$$

There is also a right-half-plane zero (F_{RHPZ}) that exists in boost mode. The frequency of the right half-plane zero is determined with Equation (42):

$$F_{RHPZ} = \frac{R_{LOAD}}{2 \times \pi \times L} \times \left(\frac{V_{IN}}{V_{OUT}}\right)^2 \quad (42)$$

The right half-plane zero increases the gain and reduces the phase simultaneously, which results in a smaller phase and gain margin. The worst-case condition occurs when the input voltage is at its minimum and the output power is at its maximum.

Current Sense Resistor Selection

The IS1N and IS1P are used for input current sense and loop control. IS2N and IS2P are used for output current sense. The series parasitic inductance (ESL) has important influence on the loop stability and constant current accuracy. **Generally, long side termination chip resistor has lower ESL than short side termination resistor.** See Figure 16 below, the left side one is the good sense resistor type.

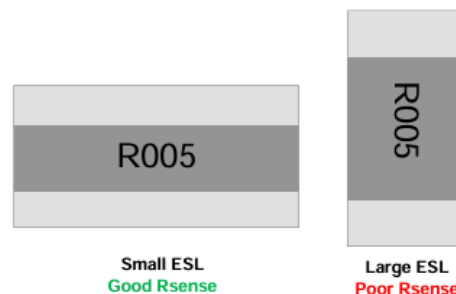


Figure 16: Recommended Rsense Type

Current sense resistor's power ratings should be 2x higher than steady state condition. 1% tolerance

MP4201应用指南

resistor is recommended for better current sense accuracy.

Several recommendations:

Rs	Part number	Power ratings
2mΩ	RL3720WT-R002-F	1W
5mΩ	RL3720WT-R005-F	
2mΩ	RL7520WT-R002-F	2W
5mΩ	RL7520WT-R005-F	

PCB Layout Guidelines

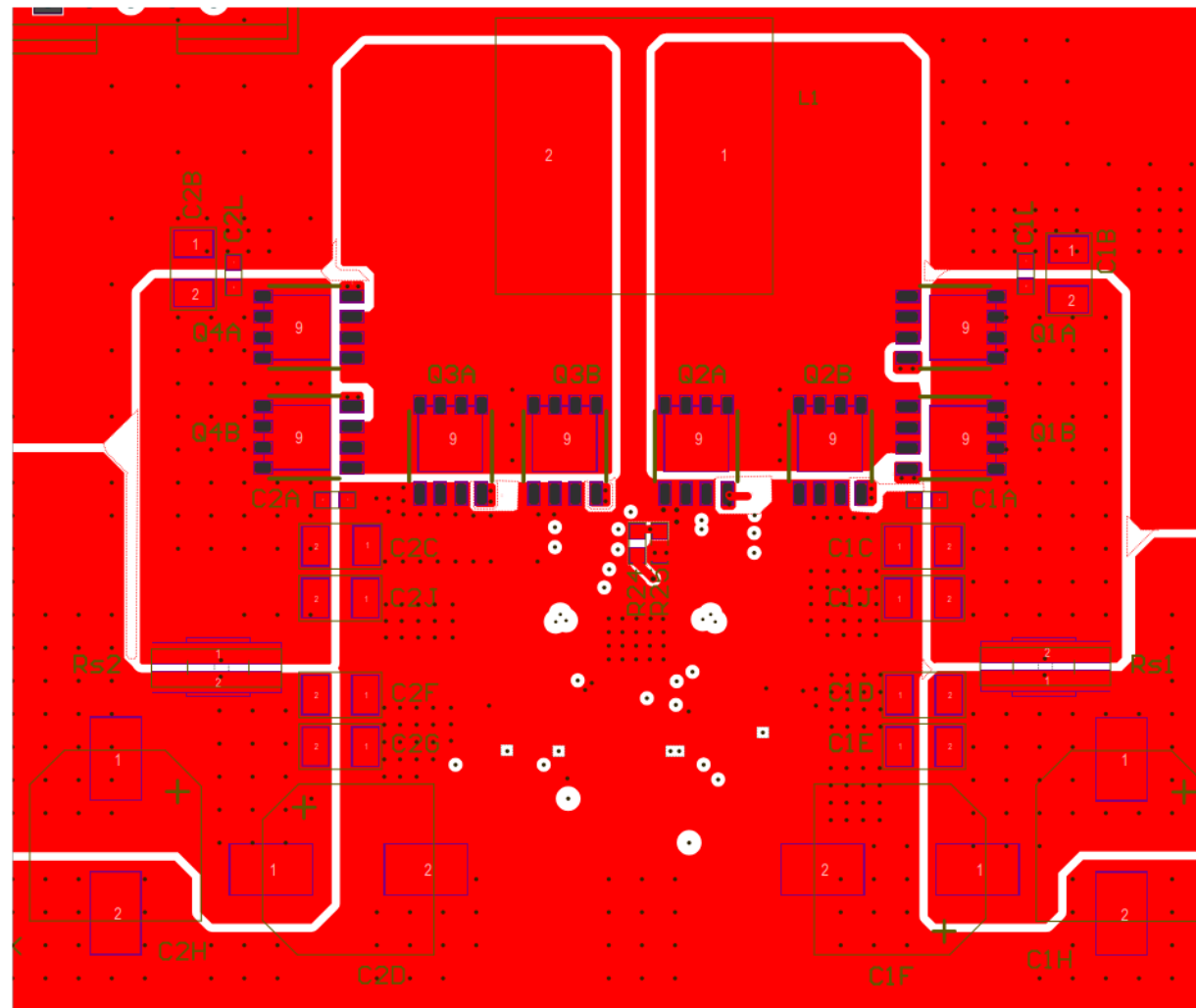
Efficient layout is a critical step in designing a buck-boost controller. Improper layout may result in reduced performance, EMI problems, resistive loss, and even system instability. For best results, refer to Figure 16 and follow the steps below:

1. In buck mode, place the input power loop — including the input filter capacitor (C_{IN}), the power MOSFETs (SWA and SWB), and the current sense resistor (RS1) — as close as possible.
2. In boost mode, place the output power loop — including the output filter capacitor (C_{OUT}), the power MOSFETs (SWC and SWD), and the current sense resistor (RS2) — as close as possible.
3. Use wide copper traces and power loop vias to help thermal dissipation.
4. Connect the exposed pad to GND, and place vias on the exposed pad for IC thermal dissipation.
5. Place small decoupling capacitors close to VIN, VOUT, and AGND.
6. Lay out the gate drive traces and return paths as directly as possible. Lay out the forward and return traces close together, either running side by side or on top of each other on adjacent layers, to minimize the inductance of the gate drive path.

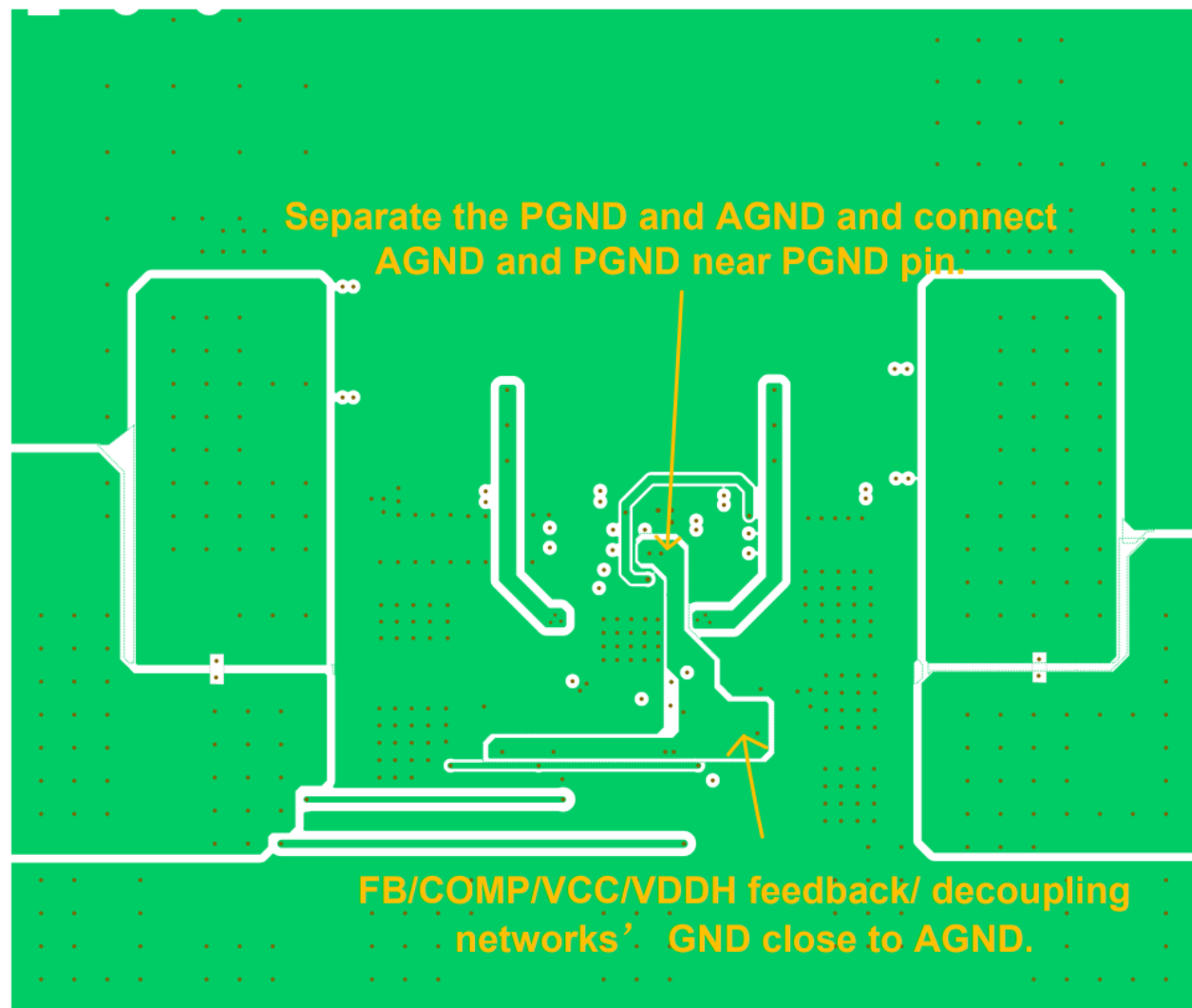
7. Use Kelvin connections to current sense resistors (Rs1 and Rs2), and run lines in parallel from the Rs1/ Rs2 terminals to the IC pins. Avoid crossing noisy areas such as SW1 and SW2 or gate drive traces.
8. Place the filter capacitor for the current sense signal as close to the IC pins as possible.
9. Place the VCC and VDDH capacitors as close as possible to the VCC and VDDH pins.
10. The feedback loop should be far away from any noise source. Place the FB dividers (R1 and R2) as close as possible to the FB and AGND pins. FB/ COMP/ VCC/ VDDH feedback/decoupling networks' GND close to AGND.
11. Separate the power and signal paths so that no power or switching current flows through the AGND connections. Connect the PGND and AGND traces near the PGND pin, near the VCC capacitor PGND connection, or near the PGND connection.
12. Place the BST1 bootstrap capacitor close to the IC, and connect directly to the BST1 and SW1 pins.
13. Place the BST2 bootstrap capacitor close to the IC, and connect directly to the BST2 and SW2 pins.

MP4201 PCB布局

Top Layer

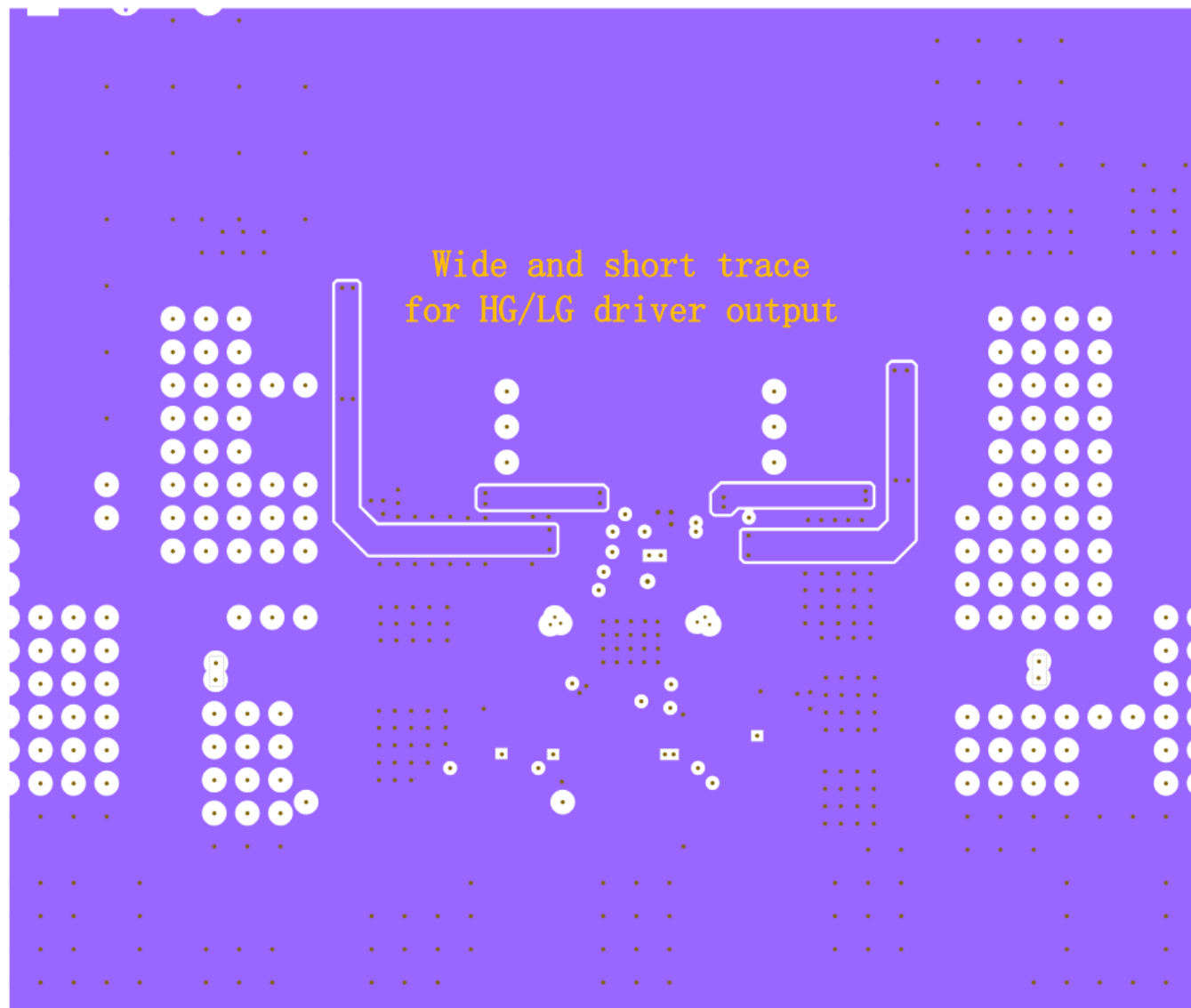


MP4201 PCB布局



Bottom Layer

MP4201 PCB布局



MID Layer 2

MP4201 PCB布局

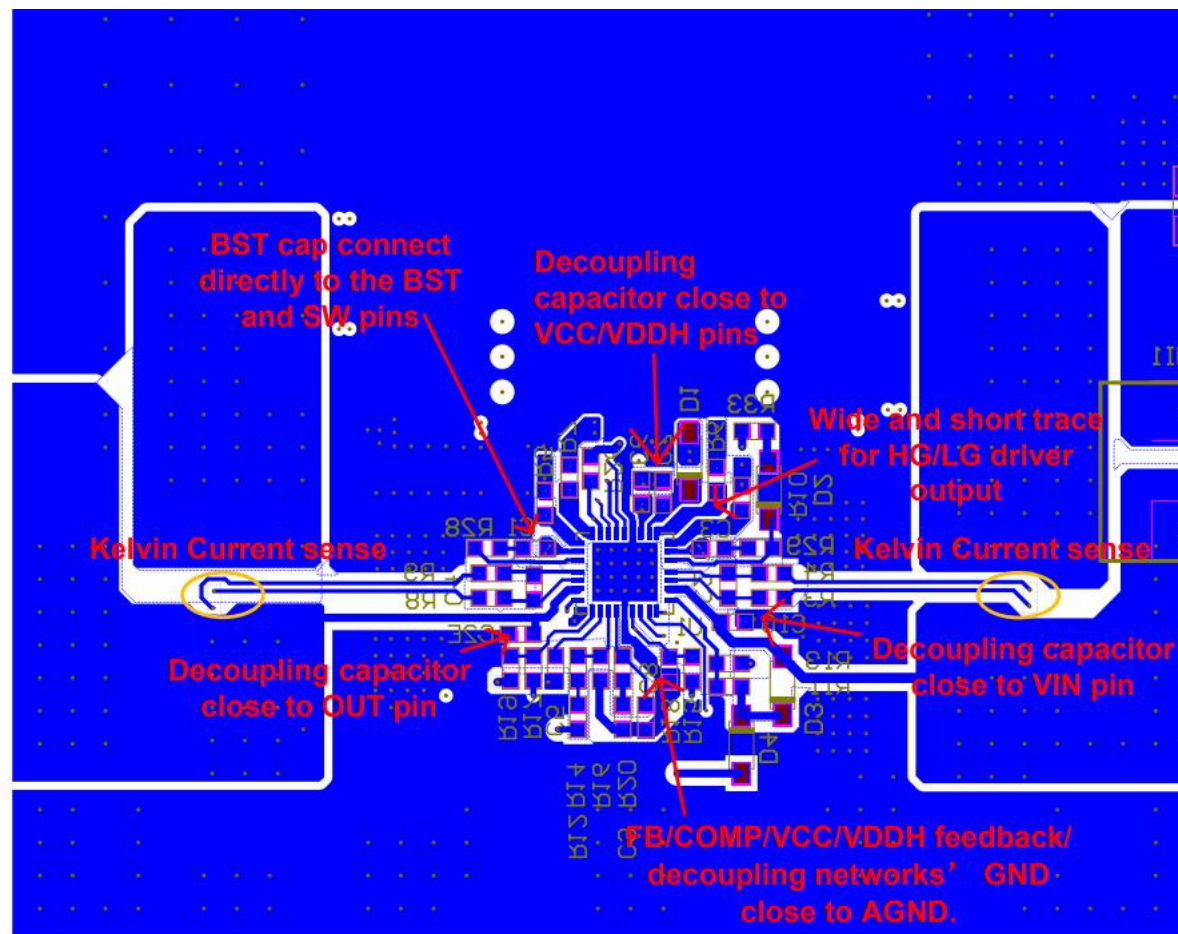


Figure 16: Recommended PCB Layout ⁽¹¹⁾

Note:

11) Parallel MOSFETs is to get higher power rating.

MP4201 典型应用电路

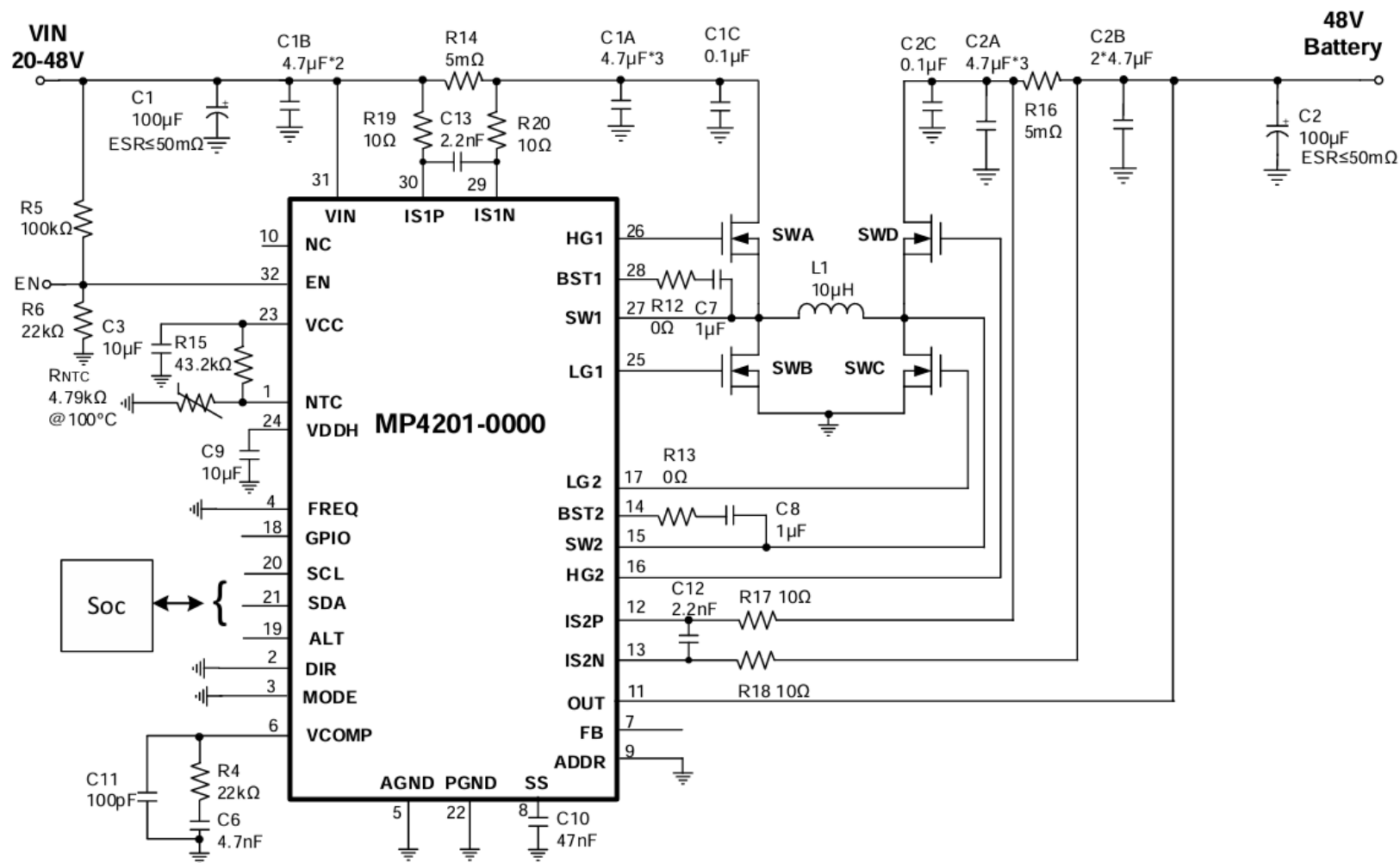


Figure 17: VIN=20V to 48V, 48V battery system, Default off, bi-direction application (12)(13)(14)

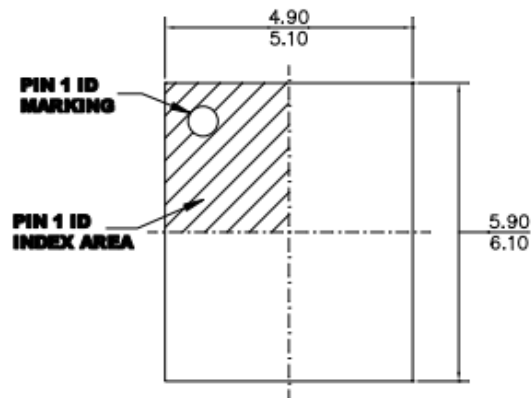
Notes:

- 12) If GaN FETs are used, recommend to series 2.2ohm resistor on all HG/LG to limit spike on SW1/2 pins .
- 13) Connect the ADDR=GND, the default I2C EN=0, the SW switching is disabled in default for MP4201-0000; connect the ADDR=VCC, the SW switching is enabled.
- 14) DIR pin and DIR register control are OR logic to determine final power direction. MCU controls the DIR register to switch power direction only when DIR pin is short to GND.

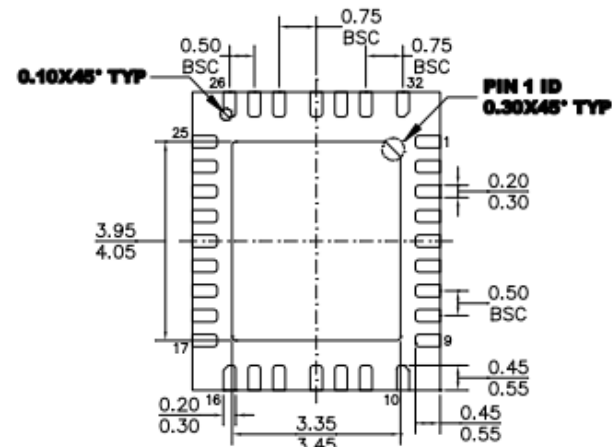
MP4201封装尺寸

PACKAGE INFORMATION

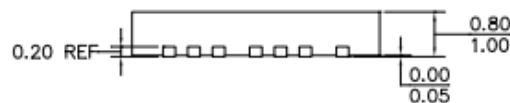
QFN-32 (5mmx6mm)



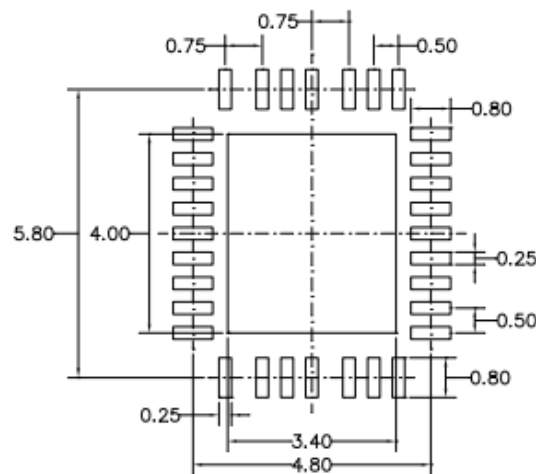
TOP VIEW



BOTTOM VIEW



SIDE VIEW

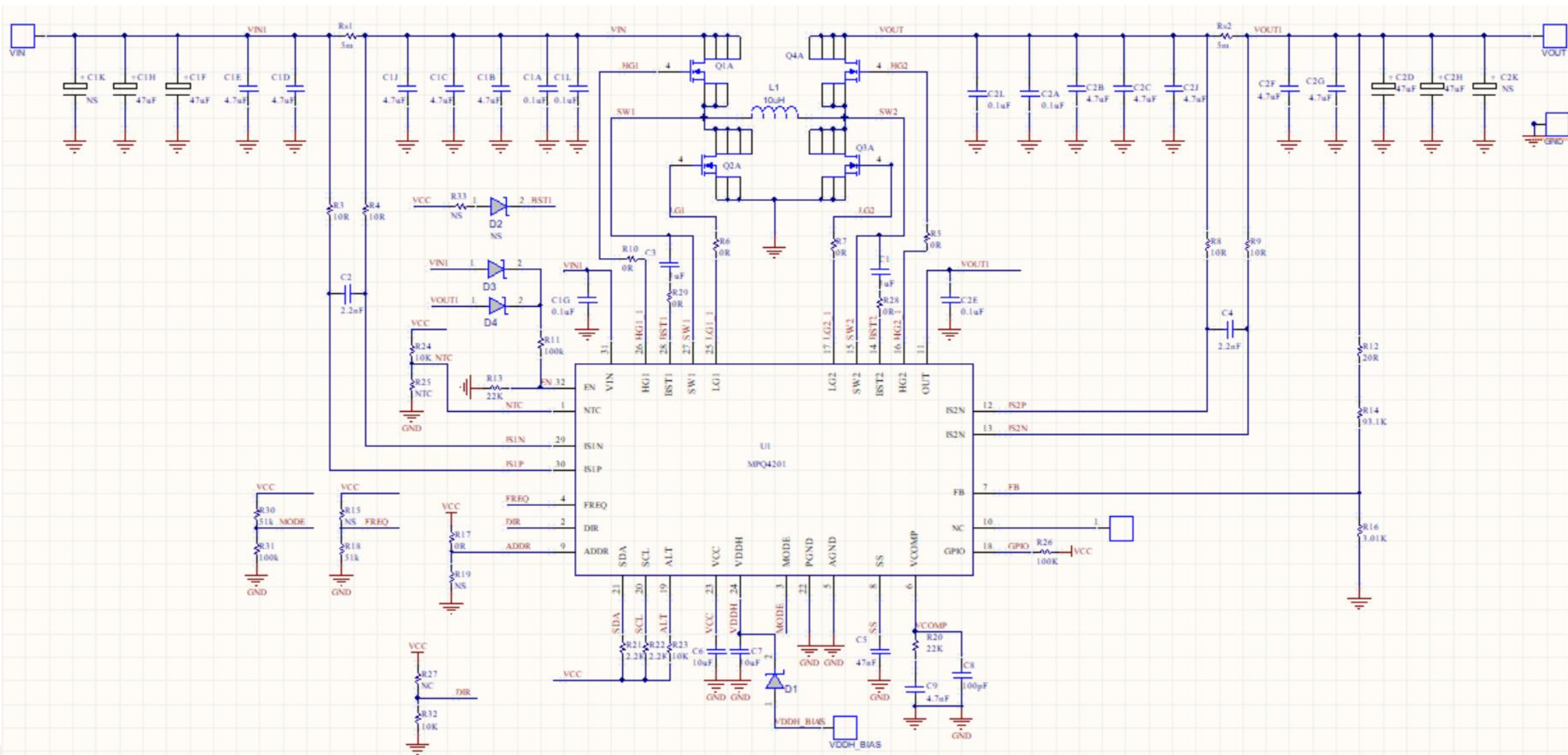


RECOMMENDED LAND PATTERN

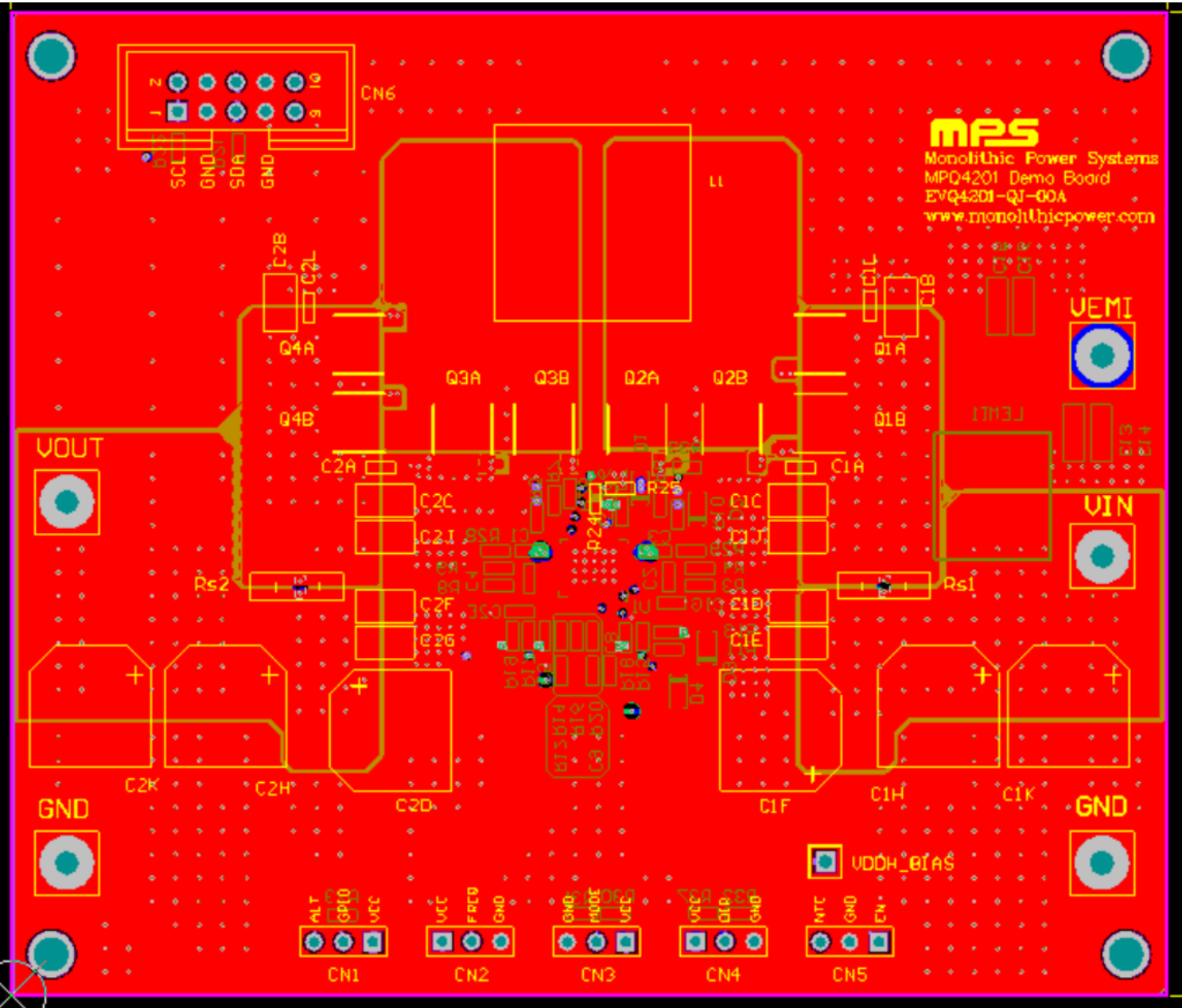
NOTE:

- 1) ALL DIMENSIONS ARE IN MILLIMETERS.
- 2) EXPOSED PADDLE SIZE DOES NOT INCLUDE MOLD FLASH.
- 3) LEAD COPLANARITY SHALL BE 0.08 MILLIMETERS MAX.
- 4) JEDEC REFERENCE IS MO-220.
- 5) DRAWING IS NOT TO SCALE.

MP4201 SCH参考设计



MP4201 PCB参考设计开源



MP4201 24V转42V 420W升压大功率
高效率电源方案 - 立创开源硬件平台



扫码了解
MP4201功率损耗计算与散热优化

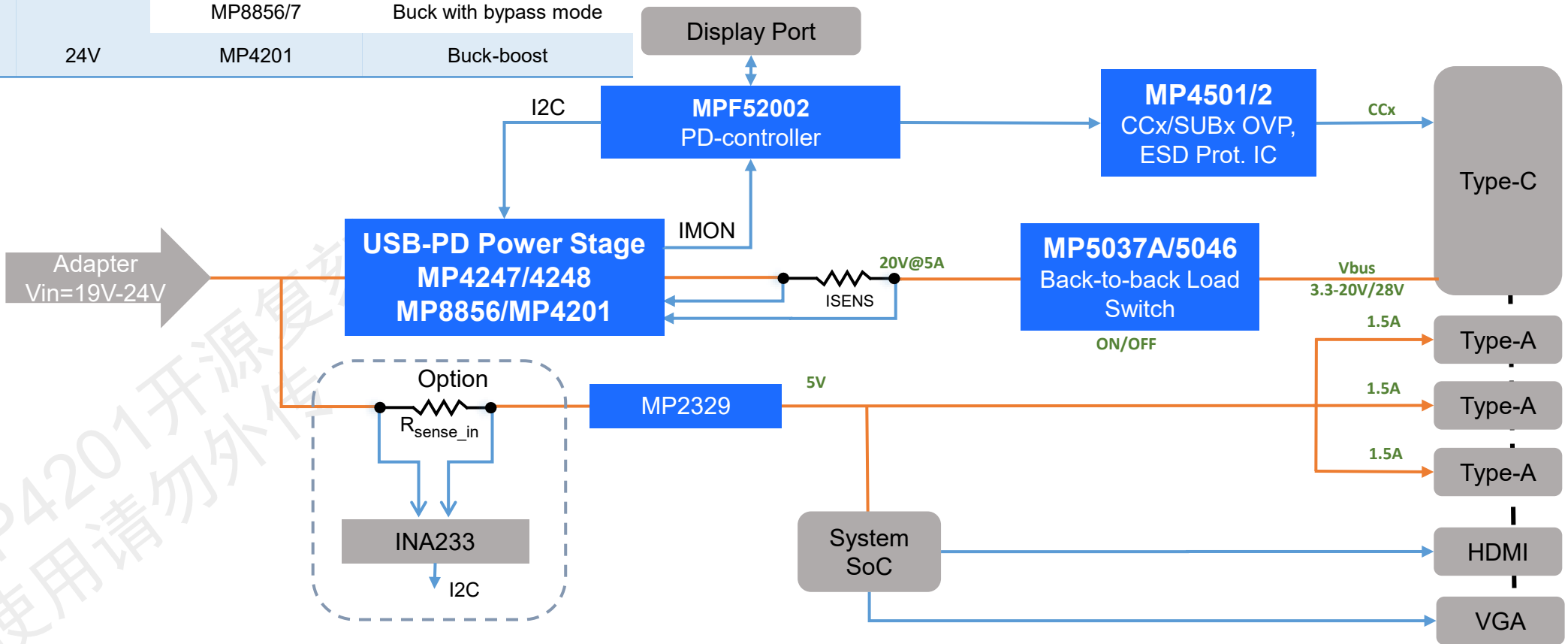


典型应用框图 | USB-C Monitor PD

MP4247/8, MP8856, MP5037A, MP4501/2 are available in Realtek 100W/140W Reference design

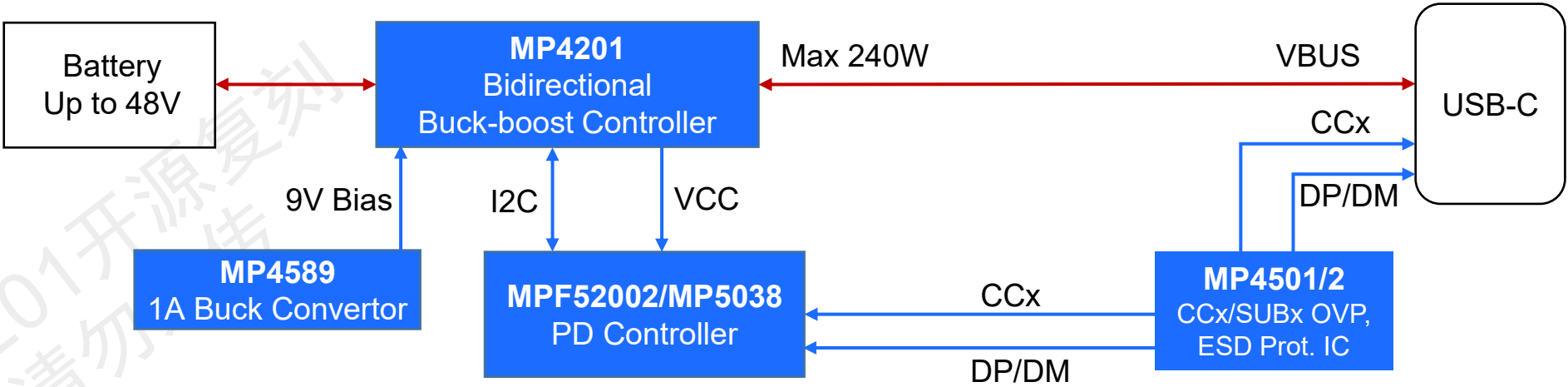
Reference Design

Pout	Vin	DCDC	Typology
65/90/140W	20V-24V	MP4248/7	Buck-boost
		MP8856/7	Buck with bypass mode
240W	24V	MP4201	Buck-boost



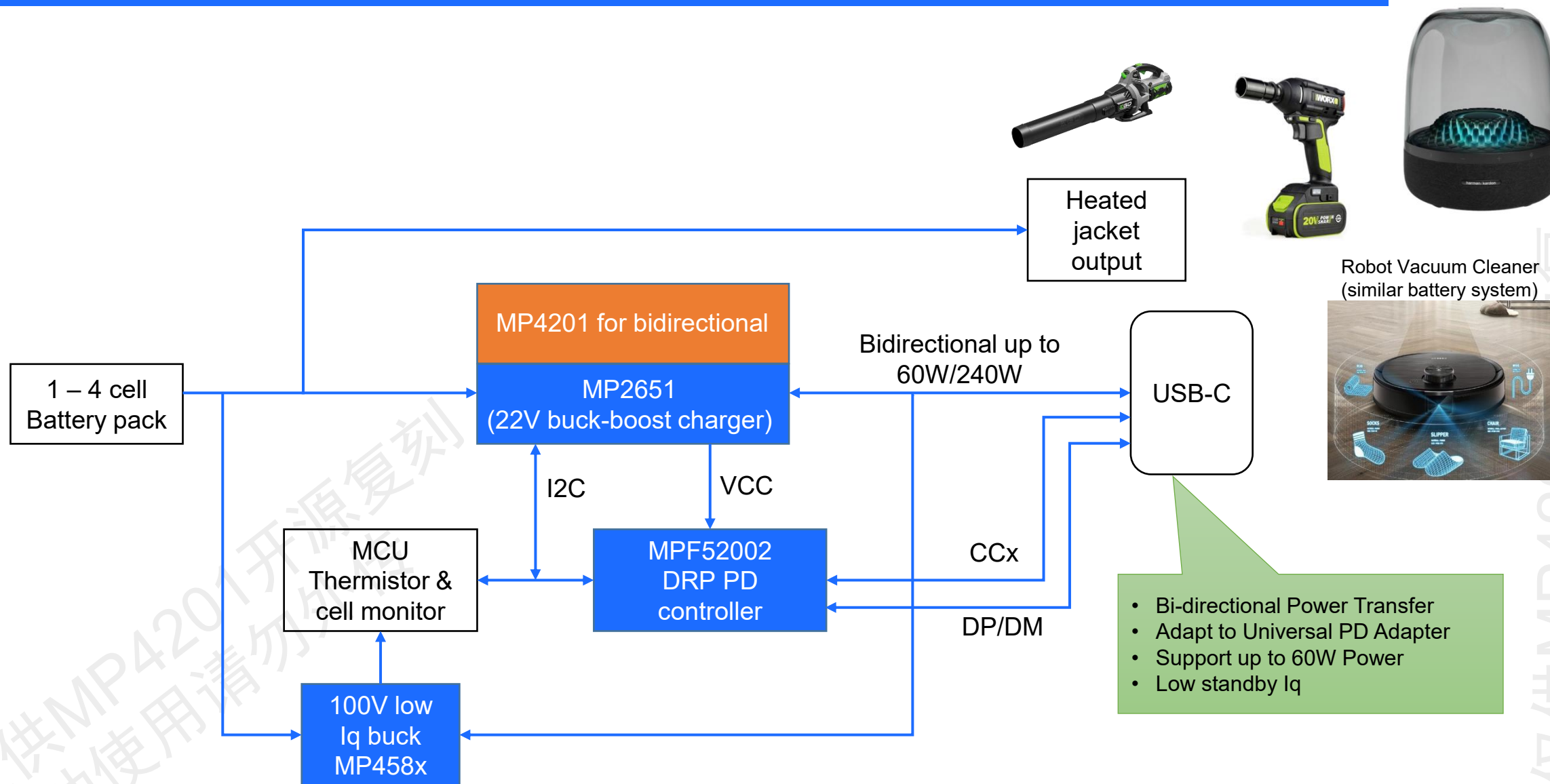
典型应用框图 | High Voltage Battery Pack PD DRP

Dir	Description
Charge	Use an external PD3.1 Adaptor to charge the battery
Discharge	Charge you cellphone by the battery



PD controller	Power	USB PD role
MP5038	<100W	DFP
MP52002	140W-240W	DRP

典型应用框图 | Sink PD / Low Voltage Battery Pack DRP



MP4201可调电源开源链接



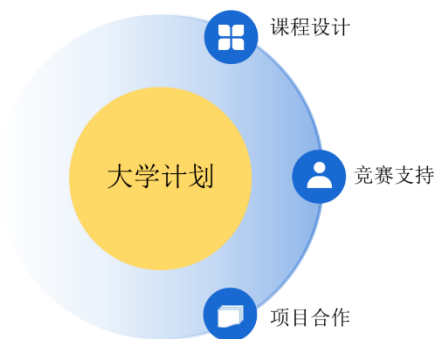
大功率可调电源-基于MP4201与MP4583设计

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